

1 Introduction

This module is concerned with solution techniques for linear ordinary and partial differential equations. These equations arise naturally in the description of physical, chemical and biological systems. The aim of the module is to make you familiar with the basic techniques and thus to enable you to study problems arising in applied mathematics. The content of the module is rather large and so we will not be able to look at things in as much detail as you might like and, in particular, there will be no attempt to be mathematically rigorous.

1.1 Definitions

Dependent and independent variable

The full solution to a differential equation, or DE, expresses the dependent variables in terms of the independent variables. So for example in

$$\frac{d^2 y}{dx^2} + 2\frac{dy}{dx} + y = 1, \quad (1)$$

y is the dependent variable, x is the independent variable and the solution $y = A \exp(-x) + Bx \exp(-x) + 1$ expresses y (the dependent variable) in terms of x (the independent variable). If a DE has more than one independent variable then it is termed a partial differential equation (PDE). An example of a PDE is

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0,$$

here ϕ is the dependent variable and x and y the independent variables. On the other hand if a DE has only one independent variable, for example (1), it is termed an ordinary differential equation (ODE).

1.1.1 Order

The order of a differential equation (DE) is that of the highest order derivative appearing in the equation. So for example

$$\frac{d^4 y}{dx^4} - \frac{d^2 y}{dx^2} = \sin(x), \quad (2)$$

is a fourth order ODE. In general an n -th order ODE can be written as

$$F(x, y(x), y'(x), \dots, y^{(n)}(x)) = 0. \quad (3)$$

1.1.2 Linearity (extremely important)

In a linear DE the dependent variable and all its derivatives appear in a linear fashion. The most general 2nd order linear ODE is thus

$$a(x)\frac{d^2y}{dx^2} + b(x)\frac{dy}{dx} + c(x)y = d(x). \quad (4)$$

Any equation for which this is not true for example

$$\frac{d^2\theta}{dt^2} = -\frac{q}{L} \sin \theta.$$

is nonlinear (**sin θ** is not linear in **θ** *i.e.*

$\sin(\alpha_1\theta_1 + \alpha_2\theta_2) \neq \alpha_1 \sin \theta_1 + \alpha_2 \sin \theta_2$).

A homogeneous equation is a linear DE in which the dependent variable appears in all terms. Hence

$$a(x)\frac{d^2y}{dx^2} + b(x)\frac{dy}{dx} + c(x)y = 0, \quad (5)$$

is the most general second order homogeneous ODE (and correspondingly (4) is non-homogeneous if $d(x) \neq 0$).

1.2 Boundary and initial conditions for ODEs

In order to find a unique solution to an ODE it is necessary to impose extra conditions on the solution. One way to do this is to specify the dependent variable and/or some of its derivatives at a particular point; this is termed imposing initial conditions. Picard's theorem states that, provided the function F and its partial derivatives are continuous in some interval containing $t = a$, the solution to the initial value problem given by n 'th order ODE (3) and the initial conditions

$$y(a) = Y_0, \quad y'(a) = Y_1, \quad \dots, \quad y^{(n-1)}(a) = Y_{n-1},$$

exists and is unique (for more details see Boyce and DiPrima). Intuitively this result is not very surprising since imposition of all the derivatives, up to the $(n - 1)$ 'th derivative, allows the n 'th derivative to be calculated at the point $x = a$ from the ODE (3). We can then differentiate this ODE once and substitute for the values of the first n derivatives to calculate the $(n + 1)$ 'th derivative; this operation can be repeated as many times as we like, and in theory we can calculate all the derivatives at the point $x = a$ and thus derive the complete Taylor series for the function.

Example. Imposing the initial conditions

$$y(0) = 2, \quad y'(0) = 3/2, \quad y''(0) = 1, \quad y'''(0) = 1/2,$$

on the fourth order equation (2) leads to the unique solution $y(x) = \sin(x)/2 + 1 + \exp(x)$.

ODEs arise from many different models of physical, chemical and

biological processes. Such models also typically give rise to conditions on the ODE. In addition to initial conditions (as discussed above) the other commonly occurring type of conditions that arises are so-called boundary conditions. These take the form of a number of conditions imposed at two different points. Usually the total number of conditions specified (at both points) is equal to the order of the differential equation. However, unlike the case in which initial conditions are specified, there is no single theorem that guarantees a unique solution.

2 Properties of linear ordinary differential equations.

The importance placed on linear differential equations in applied mathematics arises because there are a great variety of analytical

techniques that can be used to solve them. In stark contrast to this there are really very few exact techniques that can be applied to nonlinear differential equations, and usually the only option available is to solve them numerically (*i.e.* on a computer). Most of the analytical techniques used to solve linear DEs are based on the superposition principle and it is therefore vital that you understand it, and the difference between linear and nonlinear DEs.

2.0.1 The superposition principle

The superposition principle states that if $\mathbf{y} = \mathbf{u}_1(\mathbf{x})$ and $\mathbf{y} = \mathbf{u}_2(\mathbf{x})$ are two solutions to a linear homogeneous ODE (*e.g.* equation (5)) then

$$\mathbf{y} = \alpha_1 \mathbf{u}_1(\mathbf{x}) + \alpha_2 \mathbf{u}_2(\mathbf{x}) \tag{6}$$

is also a solution to the same ODE for any constants α_1 and α_2 (this rule also holds for linear homogeneous PDEs). The proof for the general homogeneous 2nd order ODE (5) is as follows (the same argument can easily be generalised to higher order ODEs). The functions $y = u_1(x)$ and $y = u_2(x)$ are solutions to (5) and hence

$$a(x)\frac{d^2u_1}{dx^2} + b(x)\frac{du_1}{dx} + c(x)u_1 = 0, \quad (7)$$

$$a(x)\frac{d^2u_2}{dx^2} + b(x)\frac{du_2}{dx} + c(x)u_2 = 0. \quad (8)$$

Adding (7) multiplied by α_1 onto (8) multiplied by α_2 gives

$$a\frac{d^2}{dx^2}(\alpha_1u_1 + \alpha_2u_2) + b\frac{d}{dx}(\alpha_1u_1 + \alpha_2u_2) + c(\alpha_1u_1 + \alpha_2u_2) = 0,$$

from which it immediately follows that $y = \alpha_1u_1 + \alpha_2u_2$ is also a solution of (5).

As it stands the superposition principle cannot be directly applied

to non-homogeneous linear DEs (*e.g.* equation (4)). However if we can find one solution to a non-homogeneous linear ODE, say $\mathbf{y} = \mathbf{y}_p(\mathbf{x})$, and a solution to the homogeneous version of the ODE $\mathbf{y} = \mathbf{u}(\mathbf{x})$ (the homogeneous version of (4) is (5)) then a second solution to (4) is $\mathbf{y} = \mathbf{y}_p(\mathbf{x}) + \mathbf{u}(\mathbf{x})$. It follows that if $\mathbf{y} = \mathbf{u}_1(\mathbf{x})$ and $\mathbf{y} = \mathbf{u}_2(\mathbf{x})$ are solutions to the homogeneous version of the ODE then

$$\mathbf{y} = \mathbf{y}_p(\mathbf{x}) + \alpha_1 \mathbf{u}_1(\mathbf{x}) + \alpha_2 \mathbf{u}_2(\mathbf{x})$$

is a solution to the non-homogeneous ODE for any constants α_1 and α_2 . The proof of this is similar to the proof of the superposition principle for homogeneous ODEs given above (Exercise: demonstrate that this is true for the general second order ODE (4)).

Example A solution of

$$y'' + y = \exp(x), \quad (9)$$

is $y = y_p(x) = \exp(x)/2$. Two linearly independent solutions of the homogeneous version of this equation $u'' + u = 0$ are

$$u = u_1(x) = \sin x \quad \text{and} \quad u = u_2(x) = \cos x.$$

It follows that

$$y = \exp(x)/2 + \alpha_1 \sin x + \alpha_2 \cos x,$$

is a solution of (9) for any constants α_1 and α_2 . In fact this is the general solution to (9).

3 Series solutions to linear ODEs

It is nearly always possible to construct solutions to a linear ordinary differential equation in terms of a series. For the sake of simplicity we shall concentrate on finding series solutions to second order homogeneous ODEs of the form

$$P(x)\frac{d^2y}{dx^2} + Q(x)\frac{dy}{dx} + R(x)y = 0. \quad (10)$$

However it is fairly straightforward to generalise the methods to non-homogeneous and higher-order ODEs. We start by looking for solutions to (10), about the point $x = x_0$, of the form

$$y = \sum_{n=0}^{\infty} a_n (x - x_0)^n, \quad (11)$$

substituting into (10), and Taylor expanding P , Q and R in powers of $(x - x_0)$. We then attempt to determine the coefficients

a_n in (11) by equating each coefficient of $x - x_0$ (*i.e.* $(x - x_0)^0$, $(x - x_0)^1$, $(x - x_0)^2$ etc.) to zero.

Example. Find a series solution to the ODE

$$(1 + x^2)y'' + xy' - y = 0, \quad (12)$$

about $x = 0$.

We look for a solution of the form

$$y(x) = \sum_{n=0}^{\infty} a_n x^n.$$

Differentiating this with respect to x gives

$$y' = \sum_{n=0}^{\infty} n a_n x^{n-1}, \quad y'' = \sum_{n=0}^{\infty} n(n-1) a_n x^{n-2}.$$

Substituting the above into (12) yields

$$\sum_{n=0}^{\infty} n(n-1)a_n x^{n-2} + n(n-1)a_n x^n + \sum_{n=0}^{\infty} n a_n x^n - \sum_{n=0}^{\infty} a_n x^n = 0.$$

After rearranging the terms this becomes

$$\sum_{n=0}^{\infty} a_n n(n-1)x^{n-2} + \sum_{n=0}^{\infty} a_n x^n (n^2 - 1) = 0.$$

Note that the x^{-1} and x^{-2} terms in the first summation disappear. We now write $n - 2 = j$ in the 1st summation and $n = j$ in the 2nd summation to find

$$\sum_{j=0}^{\infty} x^j [(j+2)(j+1)a_{j+2} + a_j(j^2 - 1)] = 0,$$

and set the coefficients in front of each power of x equal to zero to

obtain the recurrence relation

$$a_{j+2} = a_j \frac{1-j}{j+2}, \quad \text{for } j \geq 0. \quad (13)$$

It is clear that neither a_0 or a_1 are determined by the recurrence relation. This should not come as a surprise since obtaining a unique solution for $y(x)$ as the result of an initial value problem requires that we specify y and y' at some point (recall this is a second order ODE and we therefore need to specify two initial conditions). If we choose to specify them at the point $x = 0$ this give the following:

$$a_0 = y(0) \quad \text{and} \quad a_1 = y'(0),$$

and all subsequent terms in the series are determined in terms of a_0 and a_1 by the recurrence relation (13). The approach we shall adopt here, though, is to look for two linearly independent solutions for $y_1(x)$ and $y_2(x)$, since this will allow us to write

down the general solution to the ODE. Here we choose $\mathbf{a_0 = 1}$ and $\mathbf{a_1 = 0}$ in $\mathbf{y_1(x)}$ and $\mathbf{a_0 = 0}$ and $\mathbf{a_1 = 1}$ in $\mathbf{y_2(x)}$. Substitution of $\mathbf{a_0 = 1}$ and $\mathbf{a_1 = 0}$ into the recurrence relation (13) leads to the coefficients of $\mathbf{y_1}$, the first few of which are

$$a_0 = 1, \quad a_2 = \frac{a_0}{2} = \frac{1}{2}, \quad a_4 = -\frac{a_2}{4} = -\frac{1}{8} \quad a_6 = -\frac{a_4}{2} = \frac{1}{16}, \dots$$

$$a_1 = 0, \quad a_3 = 0, \quad a_5 = 0, \quad \dots, \quad a_{2n+1} = 0.$$

The series solution for $\mathbf{y_1}$ is thus

$$\begin{aligned} y_1(x) &= 1 + \frac{x^2}{2} - \frac{x^4}{8} + \frac{x^6}{16} - \dots \\ &= 1 + \sum_{m=1}^{\infty} x^{2m} \frac{(3-2m)(5-2m)\dots 1}{2m(2m-2)\dots 2}. \end{aligned}$$

Substitution of $\mathbf{a_0 = 0}$ and $\mathbf{a_1 = 1}$ into the recurrence relation (13) leads to the coefficients of $\mathbf{y_2}$

$$\mathbf{a_1 = 1}$$

$$\left. \begin{array}{l} a_{2n} = 0 \\ a_{2n+1} = 0 \end{array} \right\} \text{ for } n \geq 1.$$

Note that the series terminates after one term and thus that $y_2(x) = x$.

The radius of convergence of the series solution for y_1 can be calculated in the standard manner by recalling that the series $u_0 + u_1 + u_2 + \cdots + u_n + u_{n+1} + \cdots$ converges if

$$\lim_{j \rightarrow \infty} \left| \frac{u_{j+1}}{u_j} \right| < 1,$$

and diverges if

$$\lim_{j \rightarrow \infty} \left| \frac{u_{j+1}}{u_j} \right| > 1.$$

In the preceding example the series for y_1 converges if

$$\lim_{j \rightarrow \infty} \left| \frac{a_{2j+2} x^{2j+2}}{a_{2j} x^{2j}} \right| = \lim_{j \rightarrow \infty} \left| \frac{x^2(1 - 2j)}{(2j + 2)} \right| = x^2,$$

is less than one and diverges if it is greater than one. Thus the series for y_1 converges for $|x| < 1$ and diverges for $|x| > 1$.

Remark. An important thing to note about this example is that a_1 and a_0 are undetermined. This allows us to find two linearly independent series solutions: $y_1(x)$, for which $a_0 = 1$, $a_1 = 0$, and $y_2(x)$, for which $a_0 = 0$, $a_1 = 1$. Since we are solving a second order linear homogeneous equation the general solution then takes the form

$$y = c_1 y_1(x) + c_2 y_2(x),$$

where $\mathbf{c}_1$ and $\mathbf{c}_2$ are arbitrary constants. If for some reason either of $\mathbf{a}_1$ and $\mathbf{a}_0$ is determined by the solution our method has, at least partially, failed since we cannot then find two independent solutions (more on this later).

Example. Find series solutions to

$$y'' - xy' + xy = 0, \quad (14)$$

about $x = 0$. Substituting

$$y = \sum_{n=0}^{\infty} a_n x^n,$$

and proceeding as before leads to the series

$$\sum_{n=0}^{\infty} a_n n(n-1)x^{n-2} - \sum_{n=0}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} a_n x^{n+1} = 0.$$

We now write

$j = n - 2$ in the 1st term,

$j = n$ in the 2nd term,

$j = n + 1$ in the 3rd term,

to obtain

$$\sum_{j=-2}^{\infty} a_{j+2}(j+1)(j+2)x^j - \sum_{j=0}^{\infty} ja_jx^j + \sum_{j=1}^{\infty} a_{j-1}x^j = 0.$$

Note that the first two terms in the 1st sum, and the first term in the second sum do not contribute anything. This allows us to rewrite the above as

$$2a_2 + \sum_{j=1}^{\infty} [a_{j+2}(j+1)(j+2) - ja_j + a_{j-1}] x^j = 0.$$

Equating powers of x equal to zero then leads to the recurrence relations

$$\begin{aligned}a_2 &= 0, \\a_{j+2} &= \frac{j a_j - a_{j-1}}{(j+1)(j+2)} \quad \text{for } j \geq 1.\end{aligned}$$

Setting $a_0 = 1$ and $a_1 = 0$ we obtain one solution, say $y_1(x)$, for which the first few coefficients are

$$a_0 = 1, \quad a_1 = 0, \quad a_2 = 0, \quad a_3 = -\frac{1}{6}, \quad a_4 = 0, \quad a_5 = -\frac{1}{40}.$$

While setting $a_0 = 0$ and $a_1 = 1$ gives the coefficients of a second solution, say $y_2(x)$,

$$a_0 = 0, \quad a_1 = 1, \quad a_2 = 0, \quad a_3 = \frac{1}{6}, \quad a_4 = -\frac{1}{12}, \quad a_5 = \frac{1}{40}.$$

The leading terms in the series for y_1 and y_2 are thus

$$y_0 = 1 - \frac{x^3}{6} - \frac{x^5}{40} + \frac{x^6}{180} + \cdots,$$
$$y_1 = x + \frac{x^3}{6} - \frac{x^4}{12} + \frac{x^5}{40} - \frac{x^6}{60} + \cdots.$$

Once again the general solution has the form

$$y(x) = c_1 y_0(x) + c_2 y_1(x),$$

for arbitrary c_1 and c_2 .

3.1 Ordinary points and Singular points

The equation

$$P(x)y'' + Q(x)y' + R(x)y = 0, \tag{15}$$

is said to have an *ordinary point* at $\mathbf{x} = \mathbf{x}_0$ if both of the functions $\mathbf{Q}(\mathbf{x})/\mathbf{P}(\mathbf{x})$ and $\mathbf{R}(\mathbf{x})/\mathbf{P}(\mathbf{x})$ are *analytic* at $\mathbf{x} = \mathbf{x}_0$; that is to say that they have *convergent* power series about $\mathbf{x} = \mathbf{x}_0$ (*i.e.* a Taylor series). If either one of $\mathbf{Q}(\mathbf{x})/\mathbf{P}(\mathbf{x})$ or $\mathbf{R}(\mathbf{x})/\mathbf{P}(\mathbf{x})$ are non-analytic at $\mathbf{x} = \mathbf{x}_0$ (*i.e.* do not have a Taylor series about $\mathbf{x} = \mathbf{x}_0$) then $\mathbf{x} = \mathbf{x}_0$ is said to be a *singular point*.

About an ordinary point $\mathbf{x} = \mathbf{x}_0$ of the equation (15) it is always possible to find two independent series solution of the form

$$y = \sum_{n=0}^{\infty} a_n (\mathbf{x} - \mathbf{x}_0)^n.$$

There is, however, no such guarantee about a singular point. If $\mathbf{x}_0$ is a singular point and both $(\mathbf{x} - \mathbf{x}_0)\mathbf{Q}(\mathbf{x})/\mathbf{P}(\mathbf{x})$ and $(\mathbf{x} - \mathbf{x}_0)^2\mathbf{R}(\mathbf{x})/\mathbf{P}(\mathbf{x})$ are analytic (*i.e.* both have Taylor series about $\mathbf{x} = \mathbf{x}_0$) then the point is said to be a *regular singular point*.

Example of what goes wrong about a singular point.

The equation

$$x^2 y'' + 4xy' + (1 + x^2)y = 0, \quad (16)$$

has a singular point about $x = 0$ since neither

$$\frac{Q(x)}{P(x)} = \frac{4}{x}, \quad \text{or} \quad \frac{R(x)}{P(x)} = 1 + \frac{1}{x^2},$$

have a convergent power series about $x = 0$. Looking for a series solution, to (16), of the form $y = \sum_{n=0}^{\infty} a_n x^n$ gives

$$\sum_{n=0}^{\infty} a_n x^n (n^2 + 3n + 1) + a_n x^{n+2} = 0.$$

The first few terms in the power series are thus

$$a_0 + 5a_1 x + (11a_2 + a_0)x^2 + \cdots = 0,$$

which implies that $a_0 = a_1 = 0$. Hence $a_2 = a_3 = a_4 = a_5 = \cdots = 0$. and the method fails.

The Euler equation

$$y'' + \alpha_0 \frac{y'}{x} + \beta_0 \frac{y}{x^2} = 0, \quad (17)$$

has solutions of the form $y = x^p$ where p satisfies the quadratic

$$p^2 + (\alpha_0 - 1)p + \beta_0 = 0.$$

Provided two roots p_1 and p_2 exist to this quadratic there are two solutions to (17) and the general solution is

$$y = Ax^{p_1} + Bx^{p_2}. \quad (18)$$

3.2 Frobenius' method for regular singular points

Suppose $x = 0$ is a regular singular point of the ODE (15) we can, since $xQ(x)/P(x)$ and $x^2R(x)/P(x)$ are analytic, write

$$\begin{aligned}\frac{Q(x)}{P(x)} &= \frac{1}{x} (\alpha_0 + \alpha_1 x + \cdots + \alpha_n x^n + \cdots), \\ \frac{R(x)}{P(x)} &= \frac{1}{x^2} (\beta_0 + \beta_1 x + \cdots + \beta_n x^n + \cdots).\end{aligned}$$

Substituting into (15) gives

$$\begin{aligned}y'' + \frac{1}{x} (\alpha_0 + \alpha_1 x + \cdots + \alpha_n x^n + \cdots) y' \\ + \frac{1}{x^2} (\beta_0 + \beta_1 x + \cdots + \beta_n x^n + \cdots) y = 0.\end{aligned}$$

The largest terms in (19) as $x \rightarrow 0$, that is the ones multiplied by the lowest powers of x , are identical to (17). It is therefore apparent that the leading (*i.e.* largest) term in the series solution for y is x^p where p satisfies the quadratic (18). This suggests that the series solutions for y , a solution to (15), take the form

$$y = \sum_{n=0}^{\infty} a_n x^{n+p},$$

about the regular singular point $x = 0$.

Example 1 The equation

$$2xy'' + (3 - 2x)y' + y = 0, \tag{19}$$

has a singular point about $x = 0$, since

$Q(x)/P(x) = 3/(2x) - 1$ and $R(x)/P(x) = 1/(2x)$ are both non-analytic (don't have convergent power series) about $x = 0$.

Furthermore $x = 0$ is a regular singular point since $xQ(x)/P(x) = 3/2 - x$ and $x^2R(x)/P(x) = x/2$ are analytic. This implies we should look for a series of the form

$$\begin{aligned} y &= \sum_{n=0}^{\infty} a_n x^{n+p} \implies y' = \sum_{n=0}^{\infty} (n+p) a_n x^{n+p-1} \\ \implies y'' &= \sum_{n=0}^{\infty} (n+p)(n+p-1) a_n x^{n+p-2}. \end{aligned} \tag{20}$$

Substituting for y in (19) yields

$$\begin{aligned} \sum_{n=0}^{\infty} (2a_n(n+p)(n+p-1)x^{n+p-1} + 3a_n(n+p)x^{n+p-1}) \\ + \sum_{n=0}^{\infty} (a_n x^{n+p} - 2(n+p)a_n x^{n+p}) = 0. \end{aligned}$$

Substituting $j = n - 1$ in the 1st summation and $j = n$ in the 2nd gives

$$\sum_{j=-1}^{\infty} (2(j+p+1)(j+p) + 3(j+p+1)) a_{j+1} x^{j+p} \\ + \sum_{j=0}^{\infty} (1 - 2(j+p)) a_j x^{j+p} = 0.$$

The indicial equation is obtained by equating the lowest power of x to zero (*i.e* the 1st term in the 1st summation); it is

$$2p^2 + p = 0 \implies p = 0 \quad \text{or} \quad p = -\frac{1}{2}.$$

Equating higher powers of x to zero in (21) gives the recurrence relation

$$a_{j+1} = \frac{2(j+p) - 1}{2(j+p+1)(j+p) + 3(j+p+1)} a_j$$

$$\implies a_{j+1} = \frac{2(j+p)-1}{(j+p+1)(2j+2p+3)} a_j$$

The 2 cases are thus

$$\begin{aligned} p = 0 : \quad a_{j+1} &= \frac{2j-1}{(j+1)(2j+3)} a_j, \\ p = -\frac{1}{2} : \quad a_{j+1} &= \frac{2j-2}{(2j+1)(j+1)} a_j. \end{aligned}$$

and the first few terms are given by

$$\begin{aligned} p = 0 : \quad a_1 &= -\frac{a_0}{3}, \quad a_2 = \frac{a_1}{10} = -\frac{a_0}{30} \\ p = -\frac{1}{2} : \quad a_1 &= -2a_0, \quad a_2 = 0, \quad a_3 = a_4 = \dots = 0. \end{aligned}$$

The series representation for the 2 solutions for y are thus

$$y_1 = a_0 \left(1 - \frac{x}{3} - \frac{x^2}{30} + \dots \right),$$

$$y_2 = a_0 x^{-1/2} (1 - 2x)$$

Example 2 The equation

$$(x^2 - 2x)y'' - y' - 2y = 0, \quad (21)$$

has a regular singular point at $x = 0$ (exercise: show this is true). We hence apply Frobenius' method and look for solutions, to (21), of the form (20); this gives

$$\sum_{n=0}^{\infty} a_n ((n+p)^2 - (n+p) - 2) x^{n+p} \quad (22)$$

$$- \sum_{n=0}^{\infty} a_n (2(n+p)^2 - (n+p)) x^{n+p-1} = 0. \quad (23)$$

The lowest power of x in this series arises in the first term of the second summation and hence the indicial equation is

$$2p^2 - p = 0 \implies p = 0 \text{ or } p = \frac{1}{2}.$$

Writing $j = n$ in the 1st term of (23) and $j = n - 1$ in the 2nd term of (23) yields the series

$$\sum_{j=0}^{\infty} a_j (j + p + 1) (j + p - 2) x^{j+p} \\ - a_{j+1} (j + p + 1) (2j + 2p + 1) x^{j+p} = 0,$$

(recall that the leading term in the second series is zero because we require the indicial equation to be satisfied). Equating coefficients of x equal to zero then leads to the recurrence relation

$$a_{j+1} = \frac{j + p - 2}{2j + 1 + 2p} a_j. \quad (24)$$

In the case $p = 0$ the recurrence relation gives the following relations between successive coefficients

$$a_1 = -2a_0, \quad a_2 = -\frac{a_1}{3} = \frac{2a_0}{3}, \quad a_3 = 0, \quad a_4 = a_5 = \dots = 0.$$

The series terminates after three terms and the corresponding solution is

$$y_1 = a_0 \left(1 - 2x + \frac{2}{3}x^2 \right).$$

In the case $p = 1/2$ the recurrence relation gives the following relations between successive coefficients

$$a_1 = -\frac{3}{4}a_0, \quad a_2 = -\frac{a_1}{8} = \frac{3a_0}{32}, \quad a_3 = \frac{a_0}{128}, \dots$$

the first few terms in the corresponding series solution are thus

$$y_2 = a_0 x^{1/2} \left(1 - \frac{3}{4}x + \frac{3}{32}x^2 + \frac{1}{128}x^3 + \dots \right). \quad (25)$$

Since we have two linearly independent solutions to (21), $\mathbf{y}_1(\mathbf{x})$ and $\mathbf{y}_2(\mathbf{x})$, we can write down the general solution to (21) in terms of a linear combination of $\mathbf{y}_1(\mathbf{x})$ and $\mathbf{y}_2(\mathbf{x})$. The radius of convergence of the second series (25) can be calculated using the ratio test; thus where we write the j 'th term in the series as j_n the ratio between it and the $j + 1$ 'th term is

$$\lim_{j \rightarrow \infty} \left| \frac{u_{j+1}}{u_j} \right| = \lim_{j \rightarrow \infty} \left| \frac{a_{j+1} x^{j+3/2}}{a_j x^{j+1/2}} \right| = \lim_{j \rightarrow \infty} \left| \frac{(j - 3/2)x}{2(j + 1)} \right| = \left| \frac{x}{2} \right|.$$

It follows that the series converges for $|x| < 2$ and diverges for $|x| > 2$.

3.2.1 Breakdown of Frobenius' Method (not examined)

The standard Frobenius' method breaks down when the indicial equation (a quadratic) has only one root for p . In this instance we

can find only one solution using Frobenius' method (this happens when $\mathbf{b} = (\mathbf{a} - 1)^2/4$ and $\mathbf{p} = (1 - \mathbf{a})/2$). In this case it is helpful to return to Euler equation (17) and observe that if there is only one solution to it of the form $\mathbf{y} = \mathbf{x}^{\mathbf{p}}$ then there is always another solution of the form $\mathbf{y} = \mathbf{x}^{\mathbf{p}} \log(\mathbf{x})$. This suggests that we should look for series solutions of the form

$$\mathbf{y}_1 = \sum_{n=0}^{\infty} \mathbf{a}_n \mathbf{x}^{n+\mathbf{p}}, \quad \mathbf{y}_2 = \mathbf{y}_1(\mathbf{x}) \log(\mathbf{x}) + \sum_{n=0}^{\infty} \mathbf{b}_n \mathbf{x}^{n+\mathbf{p}}.$$

The other instance where Frobenius' method may break down is where the roots of the indicial equation ($\mathbf{p}_1$ and $\mathbf{p}_2$) differ by an integer. The problem that can occur here is that recurrence relation, relating successive coefficients $\mathbf{a}_n$, involves division by zero for some n . If $\mathbf{p}_1 - \mathbf{p}_2$ is an integer then we should look for

solutions of the form

$$y_1 = \sum_{n=0}^{\infty} a_n x^{n+p_1}, \quad y_2 = \alpha y_1(x) \log(x) + \sum_{n=0}^{\infty} b_n x^{n+p_2},$$

where $p_1 > p_2$ and α is either 0 or 1.

4 Special Functions

In this section we briefly discuss the properties of two types of special function, namely Legendre functions and Bessel functions, which occur frequently in applied mathematics and, in particular, have applications in the solution of certain linear partial differential equations. There are a lot of facts in the section and the aim is not that you should learn all of these, but rather be familiar with them and their uses (in the exam you will be provided with relevant formulae where necessary).

4.1 Legendre Functions

Legendre's equation of order α is

$$(1 - x^2)y'' - 2xy' + \alpha(\alpha + 1)y = 0. \quad (26)$$

In application α is frequently an integer and we shall mainly concentrate on this case.

Non integer α . Note that Legendre's equation (26) is invariant under the transformation $x \rightarrow -x$. It follows that if $P_\alpha(x)$ is a solution to (26) then so is $P_\alpha(-x)$. For non-integer α , $P_\alpha(-x)$ is not a multiple of $P_\alpha(x)$ (*i.e.* they are linearly independent) and the general solution to (26) is

$$y(x) = AP_\alpha(x) + BP_\alpha(-x),$$

where A and B are arbitrary constants. The functions $P_\alpha(x)$ can be plotted using Maple and their properties are tabulated in a number of books. An important property of $P_\alpha(x)$ is that it is singular at $x = -1$ for α non-integer. It follows that non-zero solutions of (26) have a singularity at, at least, one of the points $x = 1$ and $x = -1$ for α non-integer.

Series solutions. The point $x = 0$ is an ordinary point of the equation and the points $x = \pm 1$ are regular singular points. This suggests that we should look for a series solution about $x = 0$ of the form

$$y = \sum_{j=0}^{\infty} a_j x^j.$$

After some algebra this ansatz leads to the following recurrence relation:

$$a_{j+2} = \frac{(j - \alpha)(\alpha + j + 1)}{(j + 1)(j + 2)} a_j,$$

(confirm this for yourself). It follows that there are two solutions with the forms

$$\begin{aligned} y_0(x) &= 1 + \sum_{m=1}^{\infty} \left[(-1)^m x^{2m} \left(\frac{\alpha(\alpha - 2) \cdots (\alpha - 2m + 2)}{(2m)!} \right) \right. \\ &\quad \left. \times ((\alpha + 1)(\alpha + 3) \cdots (\alpha + 2m + 1)) \right], \\ y_1(x) &= x + \sum_{m=1}^{\infty} \left[(-1)^m x^{2m+1} \left(\frac{(\alpha - 1)(\alpha - 3) \cdots (\alpha - 2m + 1)}{(2m + 1)!} \right) \right. \\ &\quad \left. \times ((\alpha + 2)(\alpha + 4) \cdots (\alpha + 2m)) \right]. \end{aligned}$$

(You are not expected to remember these formulae).

Integer α . It is notable that if α is an even integer the series for $y_0(x)$ terminates when $m = \alpha/2 - 1$ and if α is an odd integer the series for $y_1(x)$ terminates when $m = (\alpha + 1)/2$. Such solutions are known as Legendre polynomials and denoted by $P_\alpha(x)$ (they occur later on in the module when we come to solve Laplace's equation by separation of variables). The first few Legendre polynomials, in a suitably normalised form, are

$$\begin{aligned} P_0(x) &= 1, \\ P_1(x) &= x, \\ P_2(x) &= \frac{1}{2}(3x^2 - 1), \\ P_3(x) &= \frac{1}{2}(5x^3 - 3x). \end{aligned}$$

In fact it turns out that the α 'th Legendre polynomial can be written as

$$P_{\alpha}(x) = \frac{1}{2^{\alpha} \alpha!} \frac{d^{\alpha}}{dx^{\alpha}} [(x^2 - 1)^{\alpha}]. \quad (27)$$

Note that these polynomials are (i) non-singular and (ii) that $P_{\alpha}(-x) = (-1)^{\alpha} P_{\alpha}(x)$. The last point means that $P_{\alpha}(x)$ and $P_{\alpha}(-x)$ are linearly dependent so that we require a second solution in order to write down the general solution to (26) for α integer. The second solution to (26), for α an integer, denoted $Q_{\alpha}(x)$, is non-polynomial and is singular at the points $x = \pm 1$. The general solution to (26), for α an integer, is thus

$$y = AP_{\alpha}(x) + BQ_{\alpha}(x),$$

where A and B are arbitrary constants.

Orthogonality. An important property of the Legendre Polynomial is orthogonality. The Legendre Polynomials $P_n(x)$ and $P_m(x)$ satisfy the equations

$$\begin{aligned} ((1 - x^2)P_n'' - 2xP_n' + n(n + 1)P_n) &= 0, \\ ((1 - x^2)P_m'' - 2xP_m' + m(m + 1)P_m) &= 0. \end{aligned}$$

Subtracting the second of these two equations, multiplied by $P_n(x)$, from the first, multiplied by $P_m(x)$, yields

$$\frac{d}{dx} ((1 - x^2)(P_mP_n' - P_nP_m')) + (n(n + 1) - m(m + 1))P_mP_n = 0,$$

Integrating the above between $x = -1$ and $x = 1$ then gives

$$\begin{aligned} &[(1 - x^2)(P_mP_n' - P_nP_m')]_{-1}^1 \\ &+ (n(n + 1) - m(m + 1)) \int_{-1}^1 P_mP_n dx = 0. \end{aligned}$$

Since P_m and P_n are non-singular at the points $x = \pm 1$ it follows

that

$$\int_{-1}^1 P_m P_n dx = 0, \quad \text{where } m \neq n \text{ or } -n - 1.$$

Where $m = n$ direct integration of the formula for the Legendre polynomial yields

$$\int_{-1}^1 P_n^2 dx = \frac{2}{2n + 1}.$$

These two relations are called *orthogonality relations* and prove very useful when constructing series solutions to certain linear PDEs. They can be summarised in the single formula as

$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2\delta_{mn}}{2n + 1} \quad m \geq 0, \quad n \geq 0. \quad (28)$$

where δ_{mn} is the Kronecker delta function.

Legendre Polynomials with negative indices.

Legendre's equation is invariant under the transformation $\alpha \rightarrow -\alpha - 1$. It follows that $P_{-\alpha-1}(x)$ is a multiple of $P_\alpha(x)$ since they both satisfy the same 2nd order ODE and are non-singular at $x = \pm 1$. It is usual, therefore, to define Legendre polynomials with negative coefficients by

$$P_{-n}(x) = P_{n-1}(x) \quad \text{for } n = 1, 2, 3 \dots$$

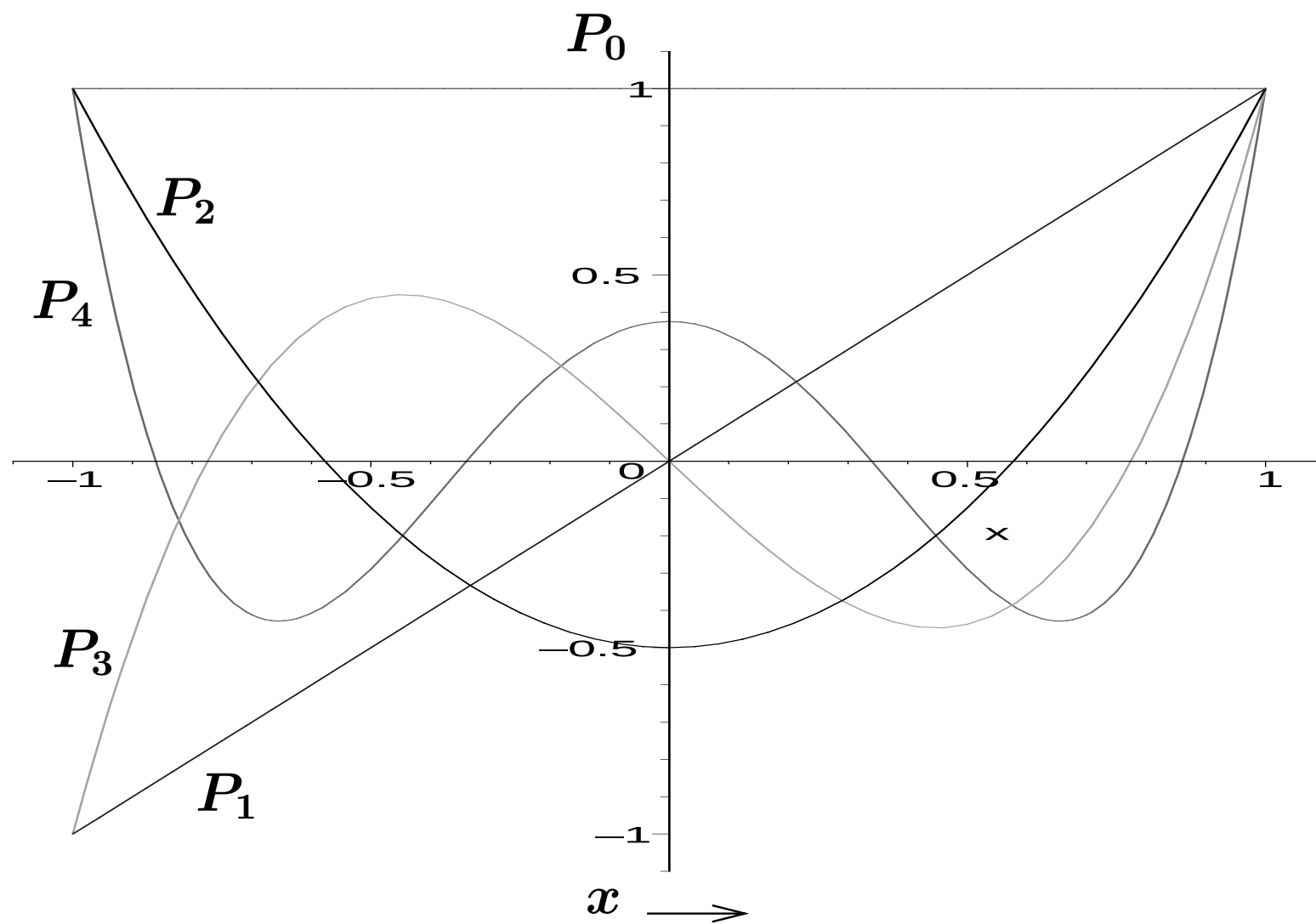


Figure 1: Plots of Legendre Polynomials.

4.2 Bessel Functions

Bessel's equation of order ν (where ν is positive) is

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0. \quad (29)$$

It has a regular singular point at $x = 0$ and this suggests that we look for a Frobenius series solution of the form

$$y = \sum_{n=0}^{\infty} a_n x^{n+p}.$$

Substitution of the above into Bessel's equation (29) yields

$$\sum_{n=0}^{\infty} [(n+p)^2 - \nu^2] a_n x^{n+p} + \sum_{n=0}^{\infty} a_n x^{n+p+2} = 0.$$

By writing $j = n$ in the first of these summations and $j = n + 2$ in the second we can obtain the series

$$a_0(p^2 - \nu^2)x^p + a_1((p+1)^2 - \nu^2)x^{p+1} + \sum_{j=2}^{\infty} [a_j((j+p)^2 - \nu^2) + a_{j-2}]x^{j+p} = 0.$$

As usual we obtain the indicial equation by setting the lowest power of x (*i.e.* x^p) equal to zero and, in this instance, this yields

$$p = \pm \nu.$$

Equating the subsequent powers of x equal to zero lead to the recurrence relations

$$a_1 = 0, \quad a_j = \frac{a_{j-2}}{\nu^2 - (j+p)^2}, \quad (30)$$

and we thus obtain two series solutions of the form

$$y_1 = x^\nu \left(1 - \frac{x^2}{4(\nu + 1)} + \frac{x^4}{32(\nu + 1)(\nu + 2)} - \cdots \right),$$

$$y_2 = x^{-\nu} \left(1 - \frac{x^2}{4(1 - \nu)} + \frac{x^4}{32(1 - \nu)(2 - \nu)} - \cdots \right),$$

unless ν is an integer. In this case, as can be seen by referring to the recurrence relation (30), the coefficient $a_{2\nu}$ is undefined when $p = -\nu$ (division by zero). However the series solution y_1 (corresponding to $p = \nu$) is still valid. If ν is an integer a second series solution may be obtained by writing

$$y_3 = y_1(x) \log(x) + x^{-\nu} \sum_{n=0}^{\infty} a_n x^n,$$

(see §3.2.1).

Rather than refer to the functions y_1 and y_2 as the generic solutions to Bessel's equations it is standard practice to refer to the

Bessel functions of the first kind J_ν and $J_{-\nu}$ which are proportional to $y_1(x)$ (*i.e.* $J_\nu(x) = Cy_1(x)$ for some constant C).

When ν is an integer a so-called Bessel function of the second kind Y_ν must be introduced. This is proportional to $y_3(x)$ (*i.e.* $J_\nu(x) = Cy_1(x)$ for some constant C). Thus the general solution to Bessel's equation (29) is

$$\begin{aligned} y &= AJ_\nu(x) + BJ_{-\nu}(x), & \text{for } \nu \text{ non-integer,} \\ y &= AJ_\nu(x) + BY_\nu(x), & \text{for } \nu \text{ integer,} \end{aligned}$$

where A and B are arbitrary constants. It is notable that both $J_{-\nu}$ and Y_ν are singular at $x = 0$. Hence if we require the solution to be non-singular at $x = 0$ (a common boundary condition) we must set $B = 0$ in both cases.

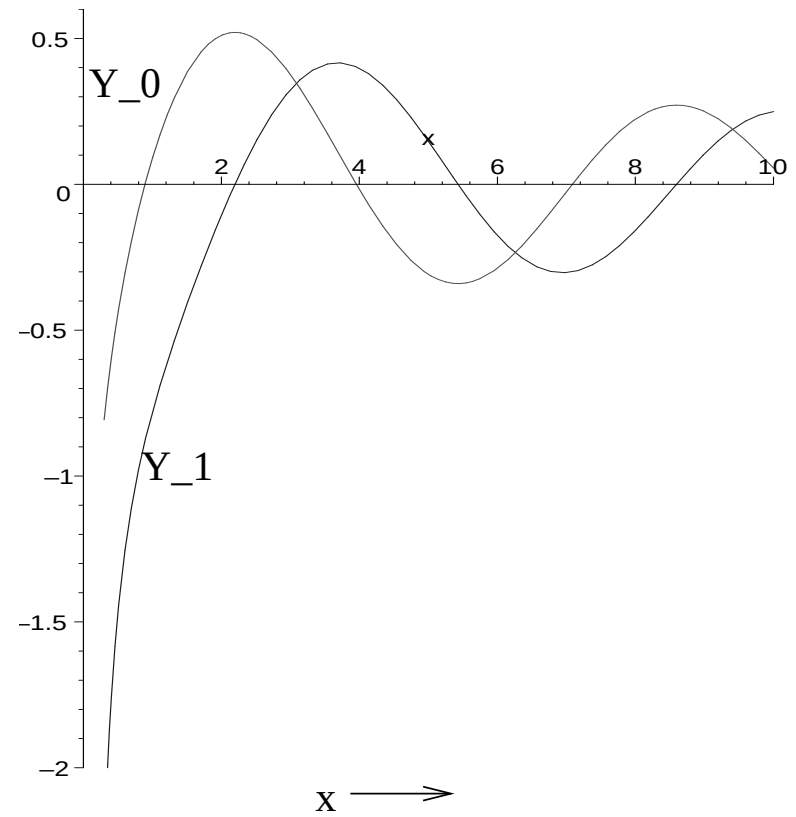
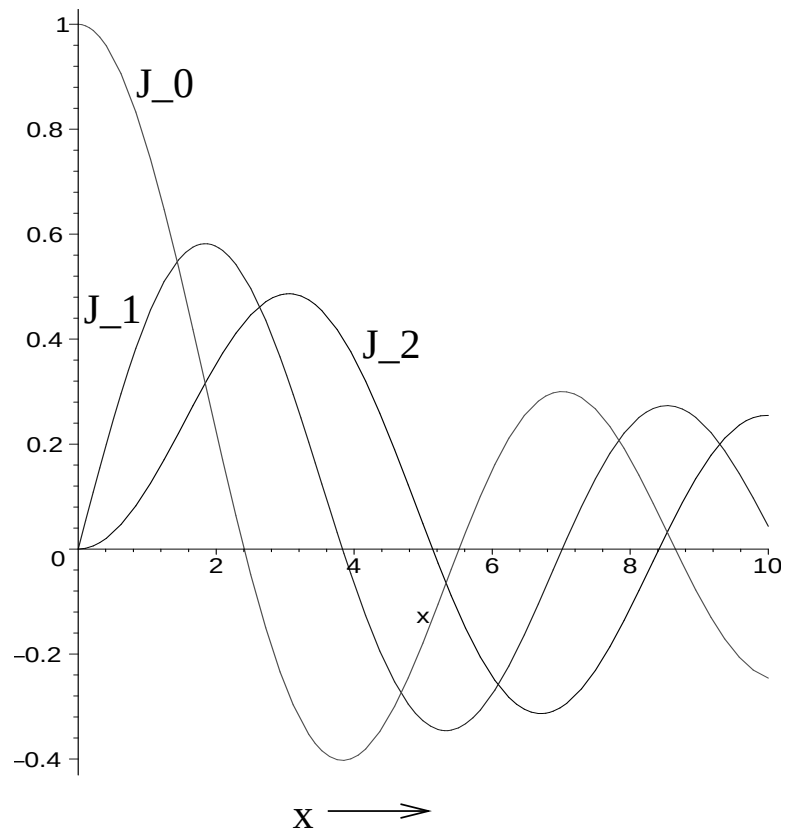


Figure 2: Plots of Bessel functions of the first and second kind.

For large x Bessel functions decay exhibiting the asymptotic behaviour

$$\left. \begin{aligned} J_\nu(x) &\sim \sqrt{\frac{2}{\pi x}} \cos \left(x - \frac{\nu\pi}{2} - \frac{\pi}{4} \right) \\ Y_\nu(x) &\sim \sqrt{\frac{2}{\pi x}} \sin \left(x - \frac{\nu\pi}{2} - \frac{\pi}{4} \right) \end{aligned} \right\} \text{ as } x \rightarrow \infty.$$

(You do not need to remember these relations).

Orthogonality. The Bessel function J_ν has infinitely many zeros (*i.e.* points λ for which $J_\nu(\lambda) = 0$). Another important property, following on from this, is the orthogonality condition. This says that if λ_m and λ_p are zeros of the Bessel function J_ν then

$$\int_0^1 x J_\nu(\lambda_m x) J_\nu(\lambda_p x) dx = 0 \quad \text{for } m \neq p, \quad (31)$$

$$\int_0^1 x J_\nu(\lambda_m x)^2 dx = \frac{1}{2} (J'_\nu(\lambda_m))^2. \quad (32)$$

We can show that (31) is true by writing $x = \lambda z$ in (29) to obtain

$$L_\lambda y = z^2 \frac{d^2 y}{dz^2} + z \frac{dy}{dz} + (\lambda^2 z^2 - \nu^2) y = 0,$$

and then writing $y_m(z) = J_\nu(\lambda_m z)$ and $y_p(z) = J_\nu(\lambda_p z)$ so that y_m and y_p satisfy the equations $L_{\lambda_m} y_m = 0$ and $L_{\lambda_p} y_p = 0$, respectively. It follows that

$$\frac{1}{z} (y_m L_{\lambda_p} y_p - y_p L_{\lambda_m} y_m) = 0,$$

and hence that

$$\frac{d}{dz} \left(z \left(y_m \frac{dy_p}{dz} - y_p \frac{dy_m}{dz} \right) \right) + (\lambda_p^2 - \lambda_m^2) z y_m y_p = 0.$$

By integrating this between 0 and 1, and noting that $y_p(1) = y_m(1) = 0$, we obtain the orthogonality condition (31),

that is

$$\int_0^1 z J_\nu(\lambda_m z) J_\nu(\lambda_p z) dz = 0 \quad \text{for } \lambda_m \neq \lambda_p.$$

Recurrence formulae for Bessel Functions. There are a number of useful formulae, relating Bessel functions of different order, which can be derived from their series representation. These include

$$\begin{aligned} J_{\nu-1}(x) + J_{\nu+1}(x) &= \frac{2\nu}{x} J_\nu(x), \\ J_{\nu-1}(x) - J_{\nu+1}(x) &= 2 \frac{d}{dx} J_\nu(x). \end{aligned}$$

(You do not need to remember these relations).

5 Periodic Functions and Fourier series

A function $f(x)$ is said to be L periodic if

$$f(x) = f(x + L) \quad \text{for any } x.$$

In this section we shall consider how such functions can be approximated using trigonometric functions. In order to simplify the analysis we shall, for the moment, restrict our attention to functions $f(\cdot)$ which are 2π periodic, such that $f(x) = f(x + 2\pi)$. With this in mind we shall attempt to approximate $f(x)$ by the 2π periodic trigonometric sum

$$f_N(x) = \frac{a_0}{2} + \sum_{n=1}^N a_n \cos(nx) + b_n \sin(nx). \quad (33)$$

Before proceeding to calculate the coefficients of this polynomial it

is helpful to evaluate the following integrals:

$$I_{mn} = \int_{-\pi}^{\pi} \cos(mx) \cos(nx) dx, \quad J_{mn} = \int_{-\pi}^{\pi} \sin(mx) \sin(nx) dx,$$

$$K_{mn} = \int_{-\pi}^{\pi} \sin(mx) \cos(nx) dx,$$

where m and n are positive integers. Using standard trigonometric identities we can write

$$I_{mn} = \int_{-\pi}^{\pi} \frac{1}{2} [\cos((m-n)x) + \cos((m+n)x)] dx,$$

$$J_{mn} = \int_{-\pi}^{\pi} \frac{1}{2} [\cos((m-n)x) - \cos((m+n)x)] dx,$$

$$K_{mn} = \int_{-\pi}^{\pi} \frac{1}{2} [\sin((m-n)x) + \sin((m+n)x)] dx.$$

It follows (for $m \geq 1$ and $n \geq 0$) that

$$I_{mn} = 0 \quad \text{for } m \neq n, \quad I_{mm} = \int_{-\pi}^{\pi} \frac{1}{2} dx = \pi,$$

$$J_{mn} = 0 \quad \text{for } m \neq n, \quad I_{mm} = \int_{-\pi}^{\pi} \frac{1}{2} dx = \pi, \quad K_{mn} = 0.$$

To summarise

$$\begin{aligned} \int_{-\pi}^{\pi} \cos(mx) \cos(nx) dx &= \pi \delta_{mn}, \\ \int_{-\pi}^{\pi} \sin(mx) \sin(nx) dx &= \pi \delta_{mn}, \\ \int_{-\pi}^{\pi} \sin(mx) \cos(nx) dx &= 0, \\ \int_{-\pi}^{\pi} 1 \cos(mx) dx &= 2\pi \delta_{0m} \\ \int_{-\pi}^{\pi} 1 \sin(mx) dx &= 0. \end{aligned} \tag{34}$$

where the Kronecker delta function δ_{mn} is equal to 1 if $m = n$ and is equal to zero otherwise.

Motivation. It is true (but we give no proof) that the functions

$$\left\{ \frac{1}{2}, \cos x, \sin x, \cos(2x), \sin(2x), \dots, \cos(nx), \sin(nx), \dots \right\}, \quad (35)$$

form a basis for the space of all piecewise continuous functions f with period 2π . In other words we can write any 2π periodic piecewise continuous function f as an infinite series of the following form:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx). \quad (36)$$

Multiplying the above by $\cos(mx)$, integrating between $-\pi$ and π

gives

$$\begin{aligned}\int_{-\pi}^{\pi} f(x) \cos(mx) dx &= \frac{a_0}{2} \int_{-\pi}^{\pi} \cos(mx) dx \\ &+ \sum_{n=1}^{\infty} \left[a_n \left(\int_{-\pi}^{\pi} \cos(mx) \cos(nx) dx \right) \right. \\ &\quad \left. + b_n \left(\int_{-\pi}^{\pi} \sin(nx) \cos(mx) dx \right) \right] \\ &= a_0 \pi \delta_{0m} + \sum_{n=1}^{\infty} a_n \pi \delta_{mn} = \pi a_m.\end{aligned}$$

Here we have made use of the results in (34). The coefficients a_m are thus

$$a_m = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(mx) dx, \quad \text{for all } m. \quad (37)$$

A similar result can be derived for the coefficients b_m by

multiplying (36) by $\sin(mx)$ and integrating between $-\pi$ and π . This gives

$$b_m = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(mx) dx, \quad \text{for all } m. \quad (38)$$

Example 1. Obtain the Fourier series for the 2π -periodic sawtooth wave

$$f(x) = \begin{cases} x + \frac{\pi}{2} & \text{for } -\pi < x < 0, \\ \frac{\pi}{2} - x & \text{for } \pi \geq x \geq 0. \end{cases}$$

Using (37) and (38) to calculate the coefficients in the Fourier series gives

$$\begin{aligned}
a_0 &= 0, \\
a_m &= \frac{1}{\pi} \left(\int_{-\pi}^{\pi} \frac{\pi}{2} \cos(mx) dx + \int_{-\pi}^0 x \cos(mx) dx \right. \\
&\quad \left. + \int_0^{\pi} -x \cos(mx) dx \right) \quad \text{for } m \geq 1, \\
b_m &= \frac{1}{\pi} \left(\int_{-\pi}^{\pi} \frac{\pi}{2} \sin(mx) dx + \int_{-\pi}^0 x \sin(mx) dx \right. \\
&\quad \left. + \int_0^{\pi} -x \sin(mx) dx \right) = 0.
\end{aligned}$$

Note that the first integrals in the expressions for both a_m and b_m are zero and that the second and third integral in the expression for a_m take the same value whilst those in the expression for b_m cancel. Thus

$$a_m = -\frac{2}{\pi} \int_0^\pi x \cos(mx) dx = -\frac{2}{\pi} \left[\frac{x \sin(mx)}{m} + \frac{\cos(mx)}{m^2} \right]_0^\pi$$

$$\implies a_m = \frac{2}{m^2 \pi} (1 - (-1)^m),$$

and the Fourier series for $f(x)$ is

$$f(x) = \sum_{m=1}^{\infty} \frac{2}{m^2 \pi} (1 - (-1)^m) \cos(mx).$$

Plots of the first 10 terms in the Fourier series are shown in figure 3.

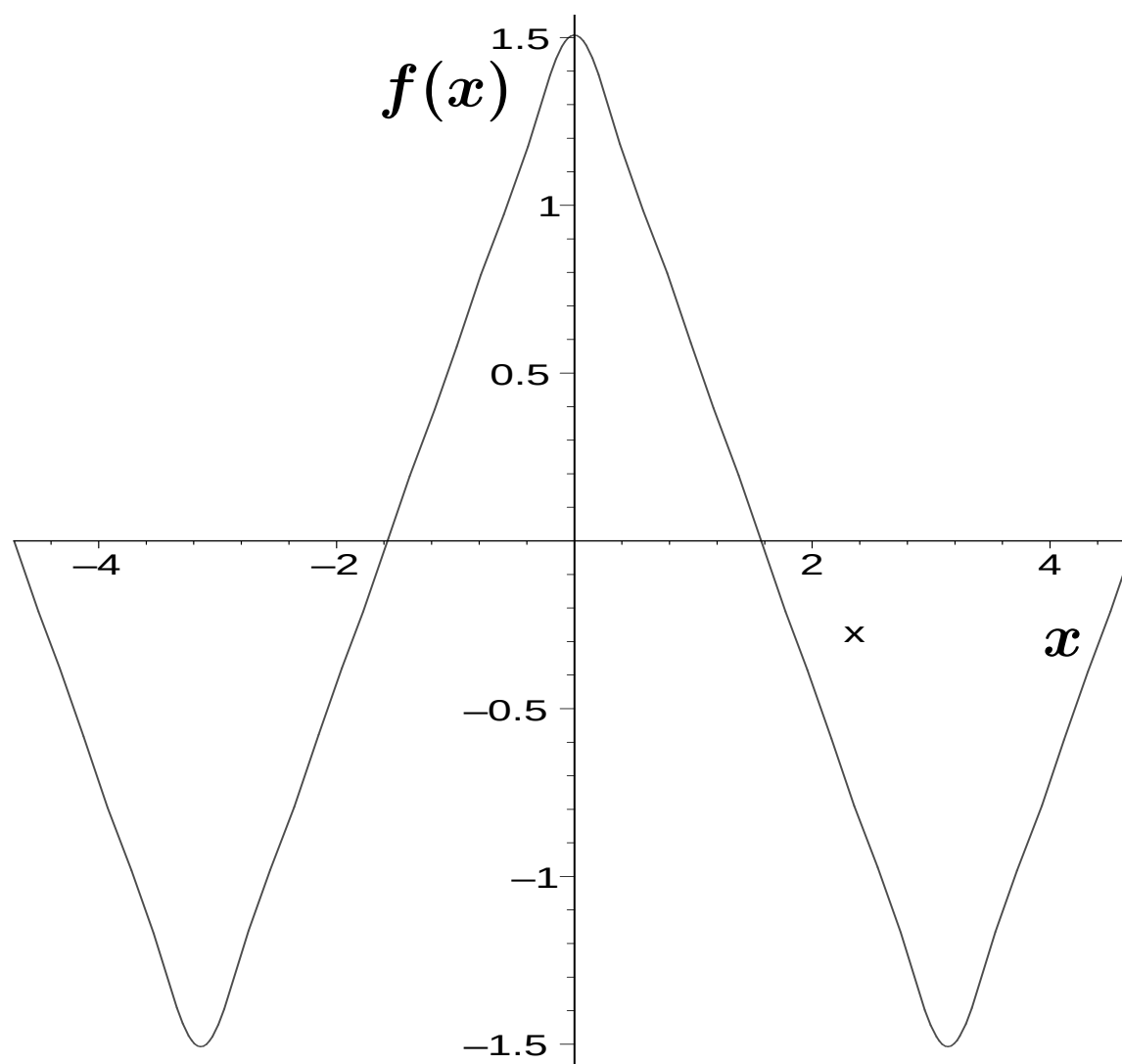


Figure 3: The first 10 terms in the Fourier series of $f(x)$.

Remark. Note that $f(x)$ is an even function and furthermore that its Fourier series contains no terms involving $\sin(mx)$. It is in general true that the Fourier series for an even function $f(x)$ takes the form

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx),$$

while that for an odd function $g(x)$ takes the form

$$g(x) = \sum_{n=1}^{\infty} b_n \sin(nx),$$

(convince yourself that these statements are true). Remembering these facts can save time when evaluating the coefficients in a Fourier series.

5.1 Fourier series for $2L$ -periodic functions

Consider now a function $f(x)$ which has period $2L$. It is easy to transform the problem of finding a Fourier series for $f(x)$ on the interval $x \in [-L, L]$ into one on the interval $[-\pi, \pi]$ by writing

$$\tilde{x} = \frac{\pi x}{L}, \quad \tilde{f}(\tilde{x}) = f(x).$$

Then the function $\tilde{f}(\tilde{x})$ has period 2π in $\tilde{x}$. It follows that its Fourier series is given by

$$\begin{aligned} \tilde{f}(\tilde{x}) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\tilde{x}) + b_n \sin(n\tilde{x}) \quad \text{where} \\ a_m &= \frac{1}{\pi} \int_{-\pi}^{\pi} \tilde{f}(\tilde{x}) \cos(m\tilde{x}) d\tilde{x}, \quad b_m = \frac{1}{\pi} \int_{-\pi}^{\pi} \tilde{f}(\tilde{x}) \sin(m\tilde{x}) d\tilde{x}. \end{aligned}$$

Transforming back to x and $f(\cdot)$ we find

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L}x\right) + b_n \sin\left(\frac{n\pi}{L}x\right), \quad (39)$$

where

$$\left. \begin{aligned} a_m &= \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{m\pi}{L}x\right) dx, \\ b_m &= \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{m\pi}{L}x\right) dx \end{aligned} \right\} \quad (40)$$

Example 2 Obtain the Fourier series for the **2**-periodic function

$$g(x) = x \quad \text{for} \quad -1 < x < 1.$$

Note that this is an odd function in x and hence $a_m = 0$ for all m . The values of b_m are determined using (40)

$$b_m = \int_{-1}^1 x \sin(m\pi x) dx = \left[-\frac{x \cos(m\pi x)}{m\pi} + \frac{\sin(m\pi x)}{(m\pi)^2} \right]_{-1}^1$$

$$\implies b_m = -\frac{2}{m\pi}(-1)^m.$$

Substituting this expression for b_m back into (39) gives the Fourier series as

$$g(x) = \sum_{n=1}^{\infty} -2\frac{1}{n\pi}(-1)^n \sin(n\pi x).$$

The first 5 and first 40 terms of this series are plotted in figure 4.

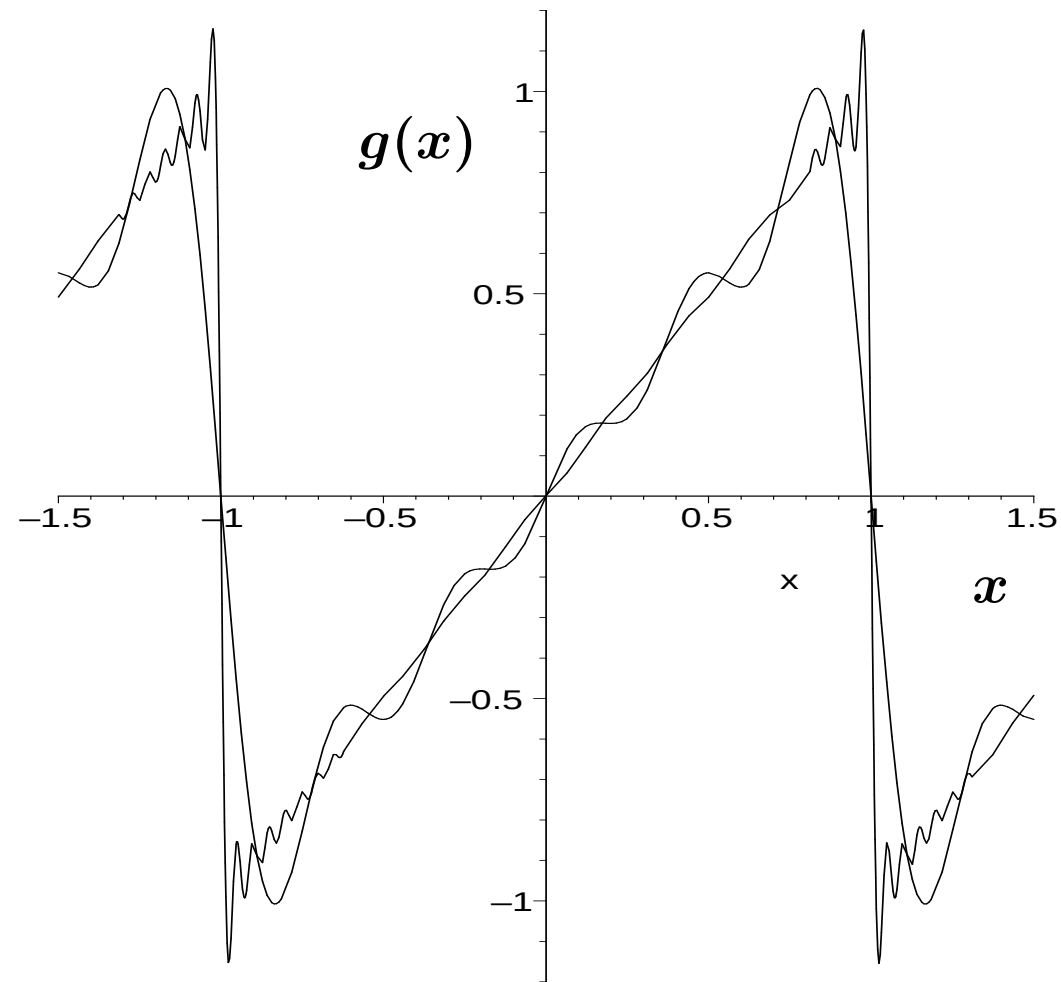


Figure 4: The first 5 terms and the first 40 terms in the Fourier series of $g(x)$.

Remark The Fourier series converged much more slowly for the function $g(x)$ (as we take more terms in the series) than it did for $f(x)$. This is because the periodic function $g(x)$ has a discontinuity in contrast to $f(x)$ which is continuous. In particular note that the Fourier series has difficulty approximating g close to the discontinuity where it has many finescale oscillations; this is termed Gibb's phenomenon.

5.2 Fourier series applied to a function defined on a finite interval

It is sometimes useful to represent a function $f(x)$ defined on a finite interval, say $x \in (0, L)$, in terms of a Fourier series. Since the function is undefined outside this interval we are free to (i) choose it to be $2L$ periodic (*i.e.* $f(x + 2L) = f(x)$) and (ii) extend it into the interval $x \in (-L, 0)$ in some manner of our

choosing. We could, for example, choose f to be even in x (*i.e.* $f(-x) = f(x)$) or odd in x (*i.e.* $f(-x) = -f(x)$).

Choosing f to be even in x implies that f can be represented by a Fourier cosine series so that

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \left(\frac{n\pi x}{L} \right), \quad (41)$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \left(\frac{n\pi x}{L} \right) dx = \frac{2}{L} \int_0^L f(x) \cos \left(\frac{n\pi x}{L} \right) dx.$$

Alternatively choosing f to be odd in x leads to a Fourier sine series representation

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \left(\frac{n\pi x}{L} \right), \quad (42)$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \left(\frac{n\pi x}{L} \right) dx = \frac{2}{L} \int_0^L f(x) \sin \left(\frac{n\pi x}{L} \right) dx.$$

In (41) and (42) we have represented the function $f(x)$ as a Fourier series in two completely different ways. Representation (41) is called a *Fourier Cosine Series* whereas (42) is termed a *Fourier Sine Series*. Note also that by choosing the periodicity of the function to be other than $2L$, or by using a different definition of $f(-x)$, we could represent the function as a Fourier series in other ways.

5.3 Solutions to periodically forced ODEs using Fourier series

It is possible to solve ODEs of the form

$$C_n \frac{d^n y}{dt^n} + C_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \cdots + C_1 \frac{dy}{dt} + C_0 y = F(t),$$

where $F(t)$ is a periodic function in t and $\{C_n, C_{n-1}, \dots, C_1, C_0\}$ are a set of constants, by use of Fourier series. In order to simplify matters we shall restrict our attention to second order ODEs of this form, that is

$$C_2 \frac{d^2 y}{dt^2} + C_1 \frac{dy}{dt} + C_0 y = F(t). \quad (43)$$

Suppose that F has period 2π (we can always rescale time so that this is true) then we can write it in the form of a Fourier series

$$F(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nt) + b_n \sin(nt)$$

We can then solve (43) by writing $y(t)$ in the form

$$y(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos(nt) + B_n \sin(nt), \quad (44)$$

and substituting this into (43) term by term. Equating coefficients of $\cos(nt)$ and $\sin(nt)$ gives

$$\left. \begin{aligned} C_0 A_0 &= a_0, \\ -C_2 n^2 A_n + C_1 n B_n + C_0 A_n &= a_n \\ -C_2 n^2 B_n - C_1 n A_n + C_0 B_n &= b_n \end{aligned} \right\} \text{for } n \geq 1.$$

We can solve these simultaneous equations for A_n and B_n to find

$$A_0 = \frac{a_0}{C_0}, \quad (45)$$

$$A_n = \frac{a_n(C_0 - C_2 n^2) - C_1 n b_n}{(C_0 - C_2 n^2)^2 + C_1^2 n^2}, \quad (46)$$

$$B_n = \frac{b_n(C_0 - C_2 n^2) + C_1 n a_n}{(C_0 - C_2 n^2)^2 + C_1^2 n^2}. \quad (47)$$

Of course the solution we have found is only a particular integral of equation (43) and in order to find the general solution we must

solve the homogeneous version of (43) (*i.e.* with $\mathbf{F} = \mathbf{0}$) to find two independent solutions. Linear combination of arbitrary multiples of these solutions with the particular integral (44) gives the general solution.

In practice it is probably easier to see what is going on by looking at a couple of examples.

Example 3 A weighted spring with natural frequency ω is forced by the the 2π periodic forcing function

$$F(t) = \begin{cases} (t + \pi) & \text{for } -\pi \leq t < 0 \\ (\pi - t) & \text{for } 0 \leq t < \pi \end{cases}$$

The vertical displacement y of the weight satisfies the equation

$$\frac{d^2 y}{dt^2} + \omega^2 y = F(t). \quad (48)$$

Find y as a function of t .

We start by expressing $F(t)$ as a Fourier series. Since r is an even function the Fourier coefficients $b_m = 0$ for all m and the coefficients a_m are given by

$$a_m = \frac{2}{\pi} \int_0^\pi (\pi - t) \cos(mt) dt \implies a_0 = \pi,$$

and

$$a_m = \frac{2}{\pi} \left[\frac{\pi}{m} \sin(mt) - \frac{1}{m^2} \cos(mt) - t \sin(mt) \right]_0^\pi$$

$$= -\frac{2}{m^2 \pi} ((-1)^m - 1), \quad m \geq 1$$

(in fact the Fourier series is the same as that in Example 1 modulo a rescaling). Substituting the Fourier series for r into (48) gives

$$\frac{d^2 y}{dt^2} + \omega^2 y = \frac{\pi}{2} - \sum_{n=1}^{\infty} \frac{2}{\pi n^2} ((-1)^n - 1) \cos(nt).$$

A particular integral of this equation can be found by looking for a

solution of the form $\mathbf{y} = \mathbf{A}_0/2 + \sum_{n=1}^{\infty} \mathbf{A}_n \cos(nt)$. Substitution into this equation reveals

$$\mathbf{A}_0 = \frac{\pi}{\omega^2}, \quad \mathbf{A}_n = \frac{2}{\pi n^2(\omega^2 - n^2)} (1 - (-1)^n).$$

Two independent solutions to the homogeneous version of (48) (*i.e.* with $\mathbf{F} = \mathbf{0}$) are $\mathbf{y}_1 = \cos(\omega t)$ and $\mathbf{y}_2 = \sin(\omega t)$ and it follows that the general solution to (48) is

$$\begin{aligned} \mathbf{y} = & \mathbf{C}_1 \cos(\omega t) + \mathbf{C}_2 \sin(\omega t) + \frac{\pi}{2\omega^2} \\ & + \sum_{n=1}^{\infty} \frac{2}{\pi n^2(\omega^2 - n^2)} (1 - (-1)^n) \cos(nt). \end{aligned} \quad (49)$$

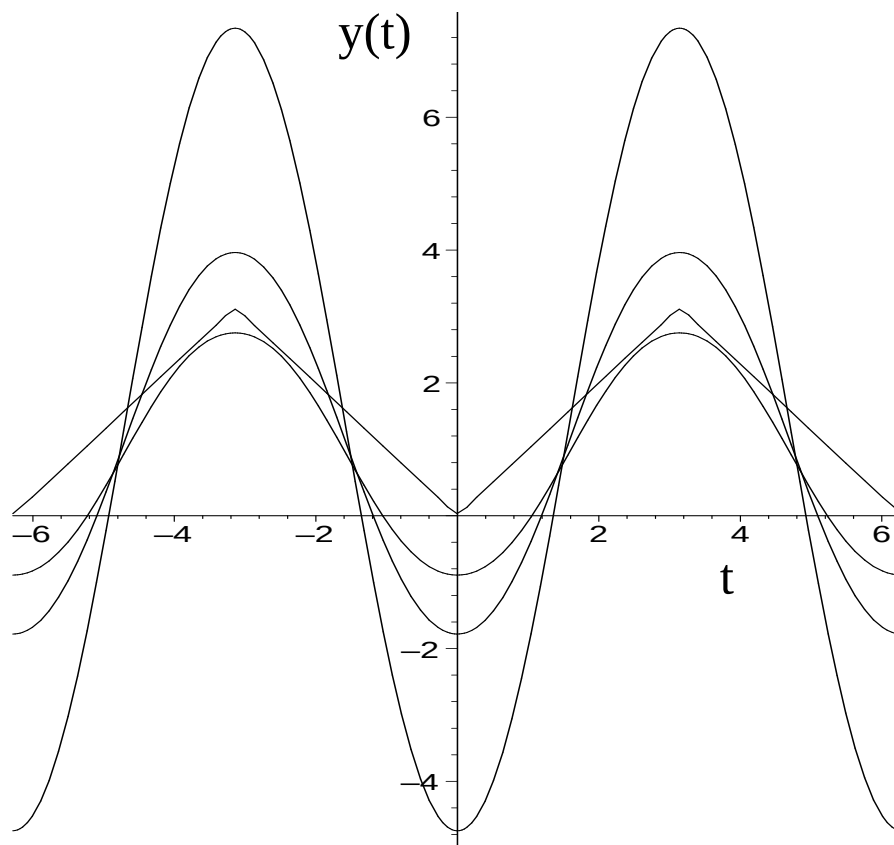


Figure 5: This picture shows $y(t)$ (with $C_1 = C_2 = 0$) for the three values of $\omega = 1.1, 1.2, 1.3$. The biggest response is for $\omega = 1.1$ and the smallest is for $\omega = 1.3$. Also shown is the forcing term in $F(t)$.

Remark Notice that as ω approaches the values $\dots, -1, 1, 3, 5, \dots$ that the response of y becomes large (see figure 5 for an example of this phenomenon). This is essentially because the expression for y (49) contains terms with divisor $\omega^2 - n^2$ (where n is an integer) and this vanishes for some n if ω is an integer. In fact if ω takes any of the values $\dots, -1, 1, 3, 5, \dots$ the solution we have found breaks down. This phenomenon is termed resonance and occur when we force a system with a forcing term that has a component of the same frequency as the natural frequency of the system. In reality this never occurs because there is always some 'damping' in physical systems. Small damping could be modelled by replacing (48) by

$$\frac{d^2y}{dt^2} + \epsilon \frac{dy}{dt} + \omega^2 y = F(t),$$

where ϵ is a small positive term. It can be seen by referring to (46) that even when ω is an odd integer we no longer run into the

problem of division by zero. That is adding a small damping term leads to large (but bounded) response even when the forcing resonates with the frequency of the system.

6 Partial Differential Equations (PDEs)

6.1 Introduction

As stated in the introduction PDEs are differential equations in which there is more than one independent variable. They arise in the modelling of a wide-range of physical phenomena including electromagnetism, fluid flow, elasticity, quantum mechanics and heat conduction and as such the ability to solve them is of considerable importance. In this module we will concentrate on the solution to linear PDE (*i.e.* ones in which the dependent variable and its derivatives appear in a linear fashion). Important examples of linear PDEs are

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0, \quad \text{Laplace's equation,} \quad (50)$$

$$\frac{\partial^2 \phi}{\partial t^2} = c^2 \frac{\partial^2 \phi}{\partial x^2}, \quad \text{The wave equation,} \quad (51)$$

$$\frac{\partial \phi}{\partial t} - k \frac{\partial^2 \phi}{\partial x^2} = f(x, t), \quad \begin{array}{l} \text{The heat equation} \\ \text{with a source term} \end{array} \quad (52)$$

An example of a nonlinear equation is given by

$$\left(\frac{\partial \phi}{\partial x} \right)^2 + \left(\frac{\partial \phi}{\partial y} \right)^2 = 1, \quad \text{The Eikonal equation}$$

As in the case of ODEs it is important to distinguish between homogeneous and non-homogeneous linear equations and the definition here is the same as that for ODEs. Thus equations (50) and (51) are homogeneous whereas equation (52) is not. The solution methods we shall adopt all rely on the superposition principle. Recall that if we have two solutions ϕ_1 and ϕ_2 to a linear

homogeneous equation (such as (50) or (51)) then this states that

$$\phi = a\phi_1 + b\phi_2.$$

is also a solution for any constants a and b . Here we will concentrate on homogeneous linear PDEs with linear (but not necessarily homogeneous) boundary/initial conditions. The superposition principle applies to homogeneous PDEs with linear homogeneous boundary/initial conditions. So, for example, we can apply the superposition principle to solutions to (51) satisfying the homogeneous set of boundary conditions

$$\phi_x|_{x=0} = 0 \quad \phi|_{x=L} = 0,$$

but not (directly) to solutions satisfying the inhomogeneous set of boundary conditions

$$\phi_x|_{x=0} = 1 \quad \phi|_{x=L} = 0.$$

It is, though, always possible to translate problems with

inhomogeneous boundary conditions into a problem with homogeneous boundary conditions and so this will not present an insurmountable difficulty.

Before describing solution methods for PDEs something must be said about initial and boundary conditions. Recall that where we solve a 2nd order ODE for u we could impose either initial conditions (e.g. $u(0) = 1$, $u'(0) = 0$) or boundary conditions (e.g. $u'(0) = 1$, $u(1) = -2$) in order to obtain a unique solution. In the case of PDEs the choice of correct initial/boundary conditions for a given equation is much more vexing, than for an ODE, and is really beyond the scope of this course. However it is such an important a point that we must say something about it. It turns out that 2nd order (quasilinear) PDEs can be classified into three classes: elliptic, hyperbolic and parabolic, exemplified by (50), (51) and (52) respectively. In order to find a unique solution to an elliptic PDE we must impose boundary data on the edge of the

domain we solve in; so, for example, if we were to solve (50) inside a circle we would need to specify ϕ (or perhaps a function of its derivatives) on the edge of the circle. In contrast 2nd order hyperbolic PDEs require 2 initial condition; thus specifying ϕ and ϕ_t at $t = 0$ (usually) gives rise to a unique solution of (51). Parabolic PDEs lie somewhere in between these two cases and require an initial condition and some boundary conditions on the side of the domain; so for example solving equation (52) with the initial and boundary conditions $\phi = \cos(\pi x)$ at $t = 0$ and $\phi_x = 0$ at $x = 0$ and $x = 1$ will lead to a unique solution.

6.2 Solution by separation of variables

In the case of a PDE with 2 independent variables (x, t) the method of separation variables consists of looking for solutions of

the form

$$\phi(x, t) = g(x)f(t). \quad (53)$$

6.2.1 The Wave equation

Suppose we wish to find a solution to the wave equation (51) subject to the homogeneous boundary conditions

$$\phi(0, t) = 0, \quad \phi(L, t) = 0. \quad (54)$$

Then we can substitute the ansatz (53) into (51) to obtain

$$f''(t)g(x) = c^2 g''(x)f(t) \implies \frac{f''(t)}{f(t)} = c^2 \frac{g''(x)}{g(x)}.$$

Observe that $f''(t)/f(t)$ is a function of t only while $c^2 g''(x)/g(x)$ is a function of x only. It follows that they must both be equal to the same constant, say $-c^2 \Lambda$. Hence f and g

satisfy the ODEs

$$(a) \quad f''(t) + c^2 \Lambda f(t) = 0, \quad (b) \quad g''(x) + \Lambda g(x) = 0. \quad (55)$$

Furthermore the boundary conditions (54) suggest we look for g satisfying the *eigenvalue* problem

$$g''(x) + \Lambda g(x) = 0, \quad g(0) = 0, \quad g(L) = 0. \quad (56)$$

Note that this is a homogeneous problem and so always has $g(x) = 0$ as a solution. Note further that Λ , the *eigenvalue*, is a free parameter which we can choose as we like. Our aim is to choose Λ so that we obtain nontrivial solutions for $g(x)$. The solutions to the ODE for g take the following forms, depending on the sign of Λ

$$g(x) = A \cos(\Lambda^{1/2} x) + B \sin(\Lambda^{1/2} x) \quad \text{for } \Lambda > 0$$

$$g(x) = A \exp((- \Lambda)^{1/2} x) + B \exp(-(- \Lambda)^{1/2} x) \quad \text{for } \Lambda < 0$$

$$g(x) = Ax + B \quad \text{for } \Lambda = 0$$

Imposing the homogeneous boundary conditions $g(0) = g(L) = 0$ gives conditions on A and B

$$\begin{aligned} \begin{pmatrix} 1 & 0 \\ \cos(\Lambda^{1/2}L) & \sin(\Lambda^{1/2}L) \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} & \Lambda > 0, \\ \begin{pmatrix} 1 & 1 \\ \exp((- \Lambda)^{1/2}L) & \exp(-(- \Lambda)^{1/2}L) \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} & \Lambda < 0, \\ \begin{pmatrix} 0 & 1 \\ L & 1 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} & \Lambda = 0, \end{aligned}$$

We can only find nontrivial solutions (*i.e* at least one non-zero A or B) if there are Λ such that the determinant of the matrix is zero.

This is only possible in the first case ($\Lambda > 0$) for $\sin(\Lambda^{1/2}L) = 0$.

That is $\Lambda^{1/2}L = \pi, 2\pi, 3\pi, \dots, n\pi, \dots$. It follows that the

solutions to (56) are

$$\Lambda^{1/2} = \frac{n\pi}{L} \quad g(x) = \sin\left(\frac{n\pi}{L}x\right), \quad n \text{ a positive integer.}$$

The corresponding solution to (55b) is

$$f(t) = C_n \cos\left(\frac{n\pi c}{L}t\right) + D_n \sin\left(\frac{n\pi c}{L}t\right).$$

and that for the wave equation (51) with the homogeneous boundary conditions (54)

$$\phi_n(x, t) = \sin\left(\frac{n\pi}{L}x\right) \left[C_n \cos\left(\frac{n\pi c}{L}t\right) + D_n \sin\left(\frac{n\pi c}{L}t\right) \right] \quad (57)$$

Since (51) and (54) are linear and homogeneous we can superpose solutions of the form (57) to obtain other solutions. The general

solution to (51) and (54) is thus

$$\phi(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{L}x\right) \left[C_n \cos\left(\frac{n\pi c}{L}t\right) + D_n \sin\left(\frac{n\pi c}{L}t\right) \right]. \quad (58)$$

The solution is still undetermined (non-unique). In order to determine it uniquely we need to prescribe initial conditions in addition to the boundary conditions (54). This motivates us to look for a solution to the initial/boundary value problem

$$\frac{\partial^2 \phi}{\partial t^2} = c^2 \frac{\partial^2 \phi}{\partial x^2}, \quad \phi(0, t) = 0, \quad \phi(L, t) = 0, \quad (59)$$

$$\phi(x, 0) = f(x), \quad \phi_t(x, 0) = g(x). \quad (60)$$

The initial conditions can be written in terms of (58), the general solution to (59)

$$\phi(x, 0) = f(x) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi}{L}x\right), \quad (61)$$

$$\phi_t(x, 0) = g(x) = \sum_{n=1}^{\infty} \left(D_n \frac{n\pi c}{L} \right) \sin \left(\frac{n\pi}{L} x \right). \quad (62)$$

We now must determine the coefficients C_n and D_n from the functions $f(x)$ and $g(x)$. Looking at (62) we note that $f(x)$ and $g(x)$ are expressed in the form of Fourier series (with coefficients C_n and $D_n n\pi c/L$, respectively). However these series are over the interval $x \in (-L, L)$, rather than the interval specified in the problem $x \in (0, L)$. Furthermore the series involve only sine terms, and are thus the series of odd functions on $(-L, L)$, (recall the discussion of Fourier sine series in the previous chapter). This suggests extending f (and g) to cover $x \in (-L, L)$ in the following manner:

$$\mathcal{F}(x) = \begin{cases} f(x) & 0 < x < L \\ -f(-x) & -L < x < 0 \end{cases} \quad \mathcal{G}(x) = \begin{cases} g(x) & 0 < x < L \\ -g(-x) & -L < x < 0 \end{cases}$$

The coefficients in the Fourier series for $\mathcal{F}$ and $\mathcal{G}$ are determined in the usual way and are

$$C_n = \frac{1}{L} \int_{-L}^L \mathcal{F}(x) \sin\left(\frac{n\pi}{L}x\right) dx = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx,$$

$$\left(D_n \frac{n\pi c}{L}\right) = \frac{1}{L} \int_{-L}^L \mathcal{G}(x) \sin\left(\frac{n\pi}{L}x\right) dx = \frac{2}{L} \int_0^L g(x) \sin\left(\frac{n\pi}{L}x\right) dx,$$

Now that we have managed to find C_n and D_n in terms of f and g we can write down the (unique) solution to the initial value problem (59)-(60); it is

$$\phi(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{L}x\right) \left[F_n \cos\left(\frac{n\pi c}{L}t\right) + \frac{L}{n\pi c} G_n \sin\left(\frac{n\pi c}{L}t\right) \right], \quad (63)$$

$$F_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx \quad G_n = \frac{2}{L} \int_0^L g(x) \sin\left(\frac{n\pi}{L}x\right) dx. \quad (64)$$

So far we have only considered solution to (51) with homogeneous

boundary conditions of the type (54). A similar method can be applied to scenarios in which other types of homogeneous boundary conditions are applied. Before doing so we will apply the method outlined above to a concrete example.

Example 1. The vertical displacement of a violin string of length ϕ satisfies the wave equation

$$\frac{\partial^2 \phi}{\partial t^2} = \frac{\partial^2 \phi}{\partial x^2}. \quad (65)$$

where x is distance along the string and t is time. The ends of the string are fixed so that

$$\phi(0, t) = 0, \quad \phi(\pi, t) = 0. \quad (66)$$

The string is plucked at time $t = 0$, starting from rest

$$\phi_t(x, 0) = 0, \quad (67)$$

from the initial position

$$\phi(x, 0) = \sin(x) + 2 \sin(3x) + \sin(4x) \quad (68)$$

Find the position of the string for times $t > 0$.

We start by looking for solutions to the homogeneous problem comprised of equation (65) and boundary conditions (66) by using separation of variables $\phi(x, t) = f(t)g(x)$. In (65) this gives

$$f''(t) + \Lambda f(t) = 0, \quad g''(x) + \Lambda g(x) = 0.$$

where Λ is an arbitrary constant. Requiring the solution to satisfy the homogeneous boundary conditions (66) leads to the following ‘eigenvalue’ problem for $g(x)$ and Λ :

$$g''(x) + \Lambda g(x) = 0, \quad g(0) = 0, \quad g(\pi) = 0.$$

This only has non-zero solutions if $\Lambda > 0$ and so we write $\Lambda = \lambda^2$;

thus

$$g(x) = A \cos(\lambda x) + B \sin(\lambda x), \quad A = 0, \quad B \sin(\lambda \pi) = 0.$$

For a non-zero solution we require $\sin(\lambda \pi) = 0$, that is $\lambda = n$ where n is an integer. There is hence a family of solutions to (65) and (66) of the form

$$\phi(x, t) = \sin(nx)(\tilde{A} \cos(nt) + \tilde{B} \sin(nt)) \quad \text{for } n \text{ an integer.} \quad (69)$$

Since (65) and (66) are homogeneous we can use the superposition principle to generate more solutions by forming linear combinations of the solutions (69). The general solution to (65) and (66) is thus

$$\phi(x, t) = \sum_{n=1}^{\infty} \sin(nx)(A_n \cos(nt) + B_n \sin(nt)) \quad (70)$$

where $\{A_n\}$ and $\{B_n\}$ are sets of arbitrary constants.

Imposing the initial condition (67) implies $B_n = 0$ for all n .

Subsequent imposition of (68) determines the values of $\{A_n\}$ and leads to the unique solution

$$\phi(x, t) = \sin(x) \cos(t) + 2 \sin(3x) \cos(3t) + \sin(4x) \cos(4t).$$

Remark. The solutions (69) describe standing waves on a string (see figure 6). Note that the frequency of the wave is $n/(2\pi)$; thus the $n = 2$ standing wave (the second harmonic) is exactly one octave above, or double the frequency of, the $n = 1$ standing wave (the first harmonic), while the $n = 3$ standing wave is a fifth above the $n = 2$ standing wave.

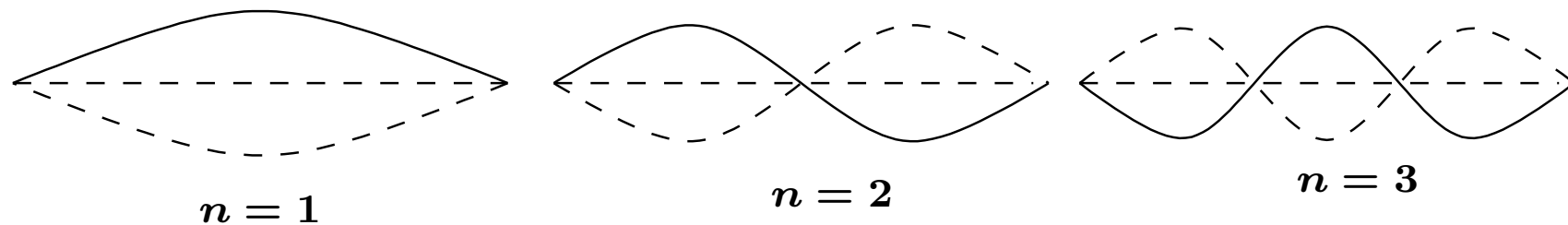


Figure 6: The first three harmonics (or standing waves) for a vibrating string with fixed ends.

Example 2 In the last example we used a rather artificial initial condition for the plucked string. More realistic initial conditions are

$$\phi(x, 0) = \begin{cases} x & \text{for } 0 < x < \pi/2 \\ \pi - x & \text{for } \pi/2 < x < \pi \end{cases} \quad (71)$$

So we replace (68) by (71) in example 1.

As in example 1 the solution satisfying (65) and (66) is (70). Here we require to determine the coefficients in (70) so that they satisfy the initial conditions (67) and (71). Initial condition (67) implies $B_n = 0$ (for all n) and (71)

$$\sum_{n=1}^{\infty} A_n \sin(nx) = \begin{cases} x & \text{for } 0 < x < \pi/2 \\ \pi - x & \text{for } \pi/2 < x < \pi \end{cases}$$

In other words we have expressed $\phi(x, 0)$ as a Fourier sine series of period 2π with coefficients A_n . In order to determine the coefficients we extend the range of $\phi(x, 0)$ to include $-\pi < x < 0$ by writing $\phi(-x, 0) = -\phi(x, 0)$, thus ensuring $\phi(x, 0)$ is an odd function. It follows that

$$A_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \phi(x, 0) \sin(nx) dx = \frac{2}{\pi} \int_0^{\pi} \phi(x, 0) \sin(nx) dx$$

$$\begin{aligned}
&= \frac{2}{\pi} \left[\int_0^{\pi/2} x \sin(nx) dx + \int_{\pi/2}^{\pi} (\pi - x) \sin(nx) dx \right], \\
&= \frac{2}{\pi} \left\{ \left[-\frac{x \cos(nx)}{n} + \frac{\sin(nx)}{n^2} \right]_0^{\pi/2} + \left[\frac{x \cos(nx)}{n} - \frac{\sin(nx)}{n^2} \right]_{\pi/2}^{\pi} \right. \\
&\quad \left. + \left[-\pi \frac{\cos(nx)}{n} \right]_{\pi/2}^{\pi} \right\}, \\
A_n &= \frac{4}{\pi} \frac{\sin(n\pi/2)}{n^2}.
\end{aligned}$$

The vertical position of the string as a function of time is thus

$$\phi(x, t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\sin(n\pi/2)}{n^2} \sin(nx) \cos(nt),$$

the position of the string between $t = 0$ and $t = \pi$ is plotted in figure 7.

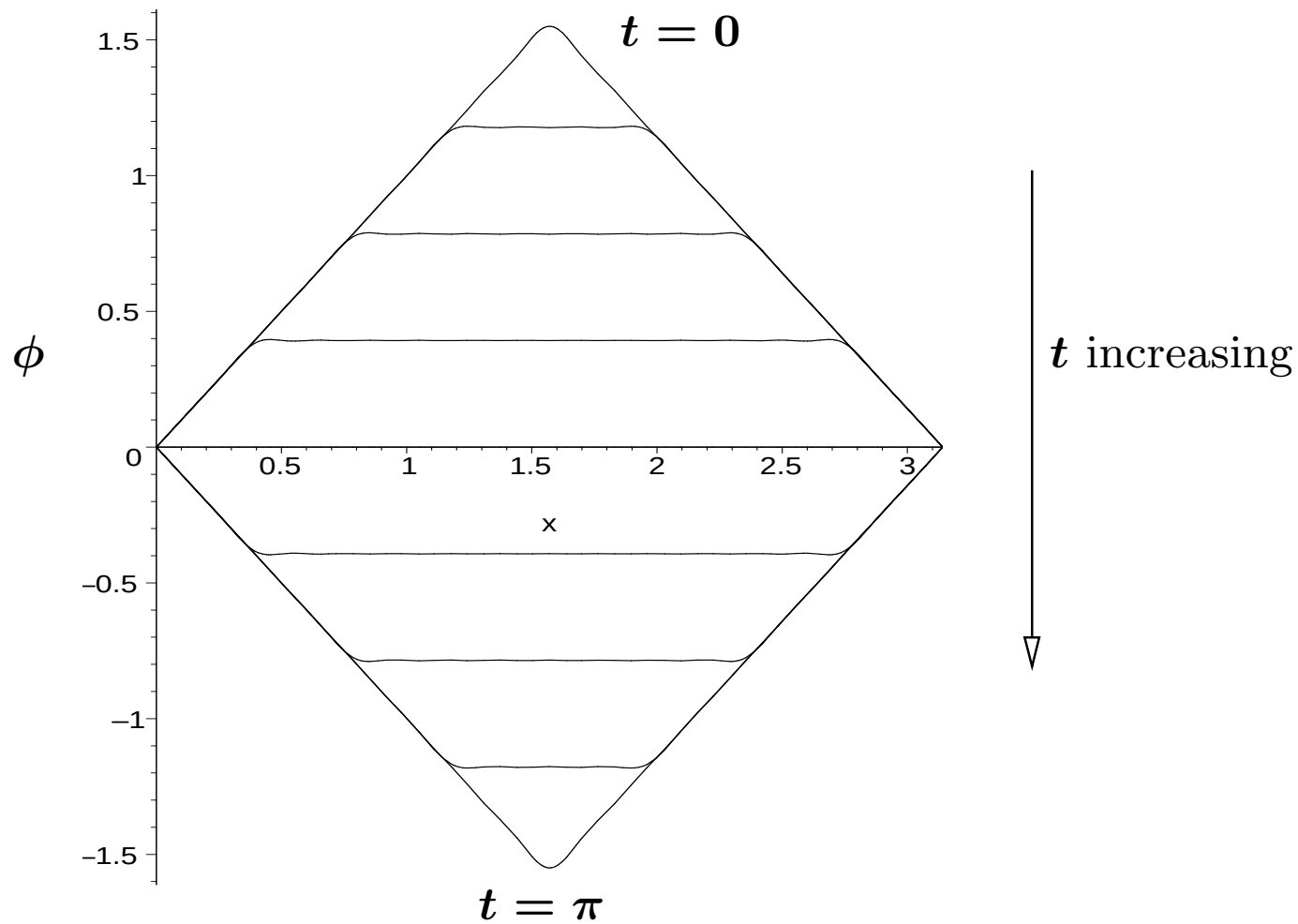


Figure 7: The position of the string, in example 2, at equally spaced intervals of $\pi/8$, between $t = 0$ and $t = \pi$.

6.2.2 The heat equation

The heat equation

$$\frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2}, \quad (72)$$

can also be tackled using the method of separation of variables since it is linear and homogeneous. This approach is illustrated in the following examples.

Example 3 A metal bar is initially at temperature $T = 1$, then at time $t = 0$, one end of the bar, $x = 1$, is heated to $T = 2$, while the other end, $x = 0$, is cooled to $T = 0$. Given that the problem can be formulated mathematically as

$$\frac{\partial T}{\partial t} = 4 \frac{\partial^2 T}{\partial x^2} \quad (73)$$

$$T(x, 0) = 1, \quad (74)$$

$$T(0, t) = 0 \quad T(1, t) = 2. \quad (75)$$

find the temperature profiles for times $t > 0$.

Boundary conditions (75) are non-homogeneous and we therefore cannot use the superposition principle directly to write down the general solution to (73) and (75). In order to circumvent this difficulty we start by subtracting off the non-homogeneous part of the boundary conditions (75). We do this by writing T as the sum of the steady solution to (73) which satisfies the boundary conditions (75), that is $T = 2x$, and an unsteady part $\theta(x, t)$

$$T = 2x + \theta(x, t).$$

This gives a homogeneous equation and boundary conditions for θ

$$\frac{\partial \theta}{\partial t} = 4 \frac{\partial^2 \theta}{\partial x^2} \quad (76)$$

$$\theta(0, t) = 0 \quad \theta(1, t) = 0. \quad (77)$$

which satisfies the initial condition

$$\theta(x, 0) = 1 - 2x. \quad (78)$$

We now use the method of separation of variables to obtain the general solution to the homogeneous problem (76)-(77). We first note that

$$\theta(x, t) = f(x)g(t),$$

is a solution to (76) if

$$f''(x) + \Lambda f(x) = 0, \quad g'(t) + 4\Lambda g(t) = 0,$$

for arbitrary Λ . In order that such a solution satisfies the homogeneous boundary conditions (77) we require Λ and f satisfy the eigenvalue problem

$$f''(x) + \Lambda f(x) = 0, \quad f(0) = 0 \quad f(1) = 0. \quad (79)$$

Nontrivial solutions to this problem can only be found for $\Lambda > 0$. Writing $\Lambda = \lambda^2$ and solving the ODE for f gives

$$f(x) = A \cos(\lambda x) + B \sin(\lambda x).$$

Applying the boundary conditions gives $A = 0$. Hence, for a nontrivial solution, we require $\sin(\lambda) = 0$. Thus $\lambda = n\pi$, for n an integer, and there are an infinite family of solutions

$$f_n(x) = B_n \sin(n\pi x) \implies \theta_n = B_n \sin(n\pi x) \exp(-4n^2\pi^2 t),$$

to the homogeneous problem (76)-(77). We can use the method of superposition to generate further solutions by linear combination.

The general solution is thus

$$\theta = \sum_{n=1}^{\infty} B_n \sin(n\pi x) \exp(-4n^2\pi^2 t). \quad (80)$$

It remains to determine the coefficients $\{B_n\}$ from the initial

conditions (78); applying this condition to (80) gives

$$\sum_{n=1}^{\infty} B_n \sin(n\pi x) = \theta(x, 0).$$

Once again we want to turn this into a Fourier series to find $\{B_n\}$. We do this by extending the range of $\theta(x, 0)$, to make it a 2 periodic function, by defining $\theta(x, 0)$ on $x \in (-1, 0)$. This is accomplished by noting that the extension of $\theta(x, 0)$ has to be an odd function in order to have a Fourier sine series. It follows that

$$\sum_{n=1}^{\infty} B_n \sin(n\pi x) = \begin{cases} -1 - 2x & \text{for } -1 < x < 0 \\ 1 - 2x & \text{for } 0 \leq x \leq 1 \end{cases}$$

The $\{B_n\}$ are then determined by the Fourier integrals

$$B_n = \left\{ \int_{-1}^0 (-2x - 1) \sin(n\pi x) dx + \int_0^1 (1 - 2x) \sin(n\pi x) dx \right\}$$

$$\begin{aligned}
&= 2 \left[\frac{2x}{n\pi} \cos(n\pi x) - \frac{2}{(n\pi)^2} \sin(n\pi x) - \frac{1}{n\pi} \cos(n\pi x) \right]_0^1, \\
B_n &= \frac{2}{n\pi} (1 + (-1)^n).
\end{aligned}$$

It follows that

$$T = 2x + \sum_{n=1}^{\infty} \frac{2}{n\pi} (1 + (-1)^n) \sin(n\pi x) \exp(-4n^2\pi^2 t).$$

The solutions is plotted in figure 8, using 80 terms in the Fourier series. T is plotted at 6 evenly spaced intervals between $t = 0$ and $t = 0.0075$. Note that the solution rapidly tends toward the steady solution $T = 2x$.

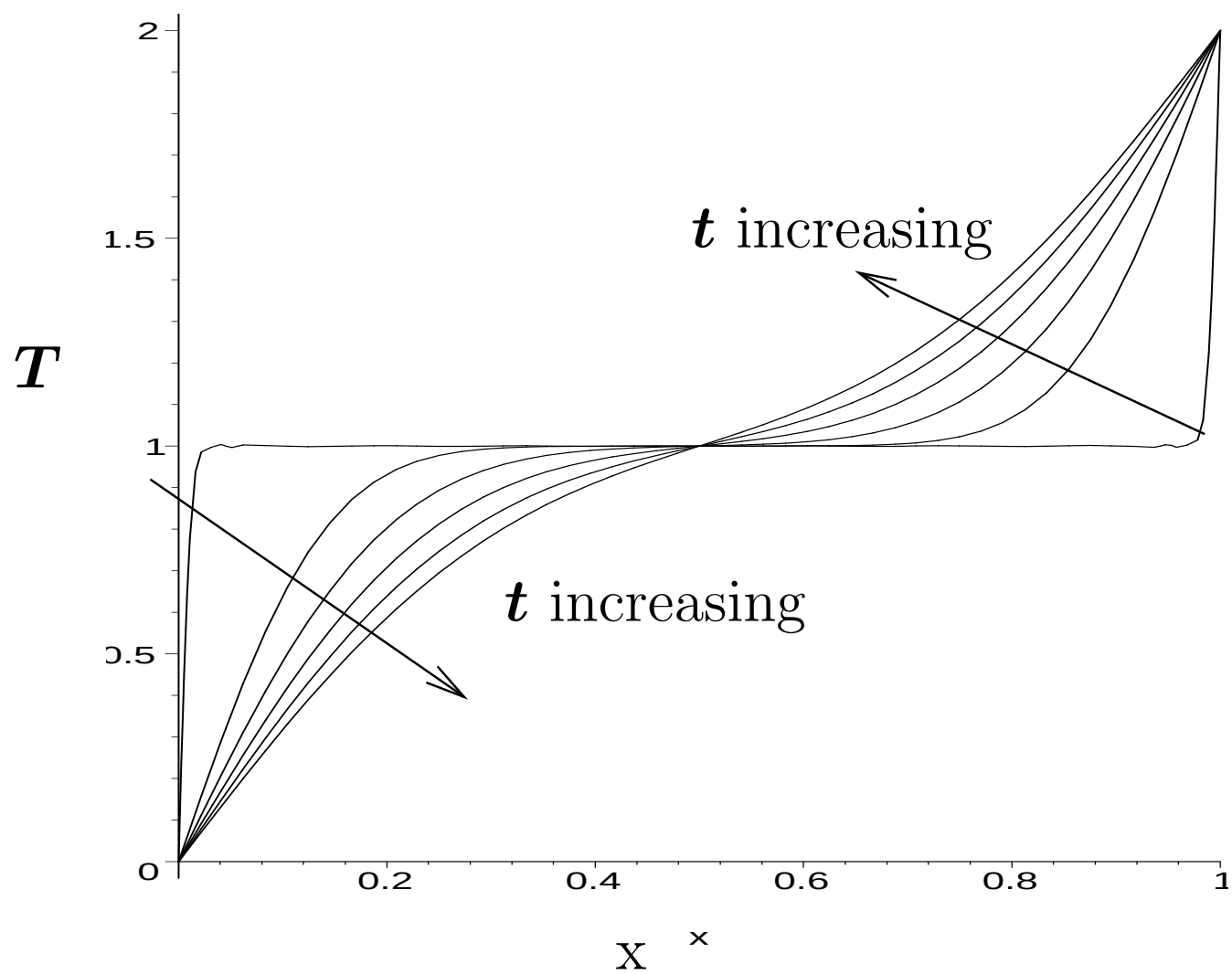


Figure 8:

6.2.3 Laplace's equation

Example 4 A stretched square membrane has height ϕ . Given the height of the membrane along the edges of the square it is possible to find its height inside the square by solving

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0, \quad (81)$$

In the instance where

$$(a) \ \phi(0, y) = 0, \quad (b) \ \phi(\pi, y) = 0, \quad (c) \ \phi(x, 0) = 0, \quad (82)$$

$$\phi(x, \pi) = G(x), \quad (83)$$

find an expression for the membrane displacement $\phi(x, y)$. Write down this expression for the special case $G(x) = x(\pi - x)$.

The method in this example follows that in the others we have tackled. We start by looking for the general solution to (81) and the homogeneous boundary conditions (82) using the separation of

variables $\phi = f(x)g(y)$. Making this substituting in (81) gives

$$(a) \quad f''(x) + \Lambda f(x) = 0, \quad (b) \quad g''(y) - \Lambda g(y) = 0. \quad (84)$$

Equation (84a) taken together with boundary conditions (82a) and (82b) comprise the eigenvalue problem

$$f''(x) + \Lambda f(x) = 0, \quad f(0) = 0, \quad f(\pi) = 0. \quad (85)$$

We can only find non-zero solutions if $\Lambda > 0$ and so write $\Lambda = \lambda^2$. With this substitution (85) reduces to

$$f(x) = A \cos(\lambda x) + B \sin(\lambda x), \quad A = 0, \quad B \sin(\lambda \pi) = 0.$$

For a nonzero solution we require $\sin(\lambda \pi) = 0$; that is λ an integer. Thus the solutions for f are

$$f(x) = B \sin(nx) \quad \text{where } n \text{ is an integer,} \quad (86)$$

The corresponding solutions to (84b) are

$$g(y) = C \sinh(n(y - D)).$$

But (82c) implies $g(0) = 0 \implies D = 0$. Putting everything together gives a family of solutions to (81) and (82)

$$\phi(x, y) = B \sin(nx) \sinh(ny).$$

The general solution to (81) and (82) is thus

$$\phi(x, y) = \sum_{n=1}^{\infty} B_n \sin(nx) \sinh(ny) \quad \text{where } n \text{ is an integer.} \quad (87)$$

It remains to determine the coefficients from the non-homogeneous boundary condition (83); this implies

$$\sum_{n=1}^{\infty} (B_n \sinh(n\pi)) \sin(nx) = G(x). \quad (88)$$

In order to find the coefficients $\{B_n\}$ we need to write (88) as a Fourier sine series. We do this by extending the range of $G(x)$ to include $x \in (-\pi, 0)$ choosing $G(x)$ to be an odd function ($G(-x) = -G(x)$) so that it has a Fourier sine series. We can then use the Fourier integrals to determine the coefficients of this Fourier series ($B_n \sinh(n\pi)$)

$$\begin{aligned} B_n &= \frac{1}{\pi \sinh(n\pi)} \left[\int_{-\pi}^0 G(x) \sin(nx) dx + \int_0^{\pi} G(x) \sin(nx) dx \right] \\ &= \frac{2}{\pi \sinh(n\pi)} \int_0^{\pi} G(x) \sin(nx) dx. \end{aligned}$$

In the particular instance where $G(x) = x(\pi - x)$

$$\begin{aligned} B_n &= \frac{4}{\pi \sinh(n\pi)} \frac{1 - (-1)^n}{n^3} \\ \implies \phi(x, y) &= \sum_{n=1}^{\infty} \frac{4(1 - (-1)^n)}{\pi n^3 \sinh(n\pi)} \sin(nx) \sinh(ny), \end{aligned}$$

(check this for yourself). A plot of the contours of $\phi(x, y)$ is made in figure 9 by using the first 20 terms in the Fourier series. The contours are plotted at intervals of **0.1** between $\phi = \mathbf{0.1}$ and $\phi = \mathbf{2.7}$.

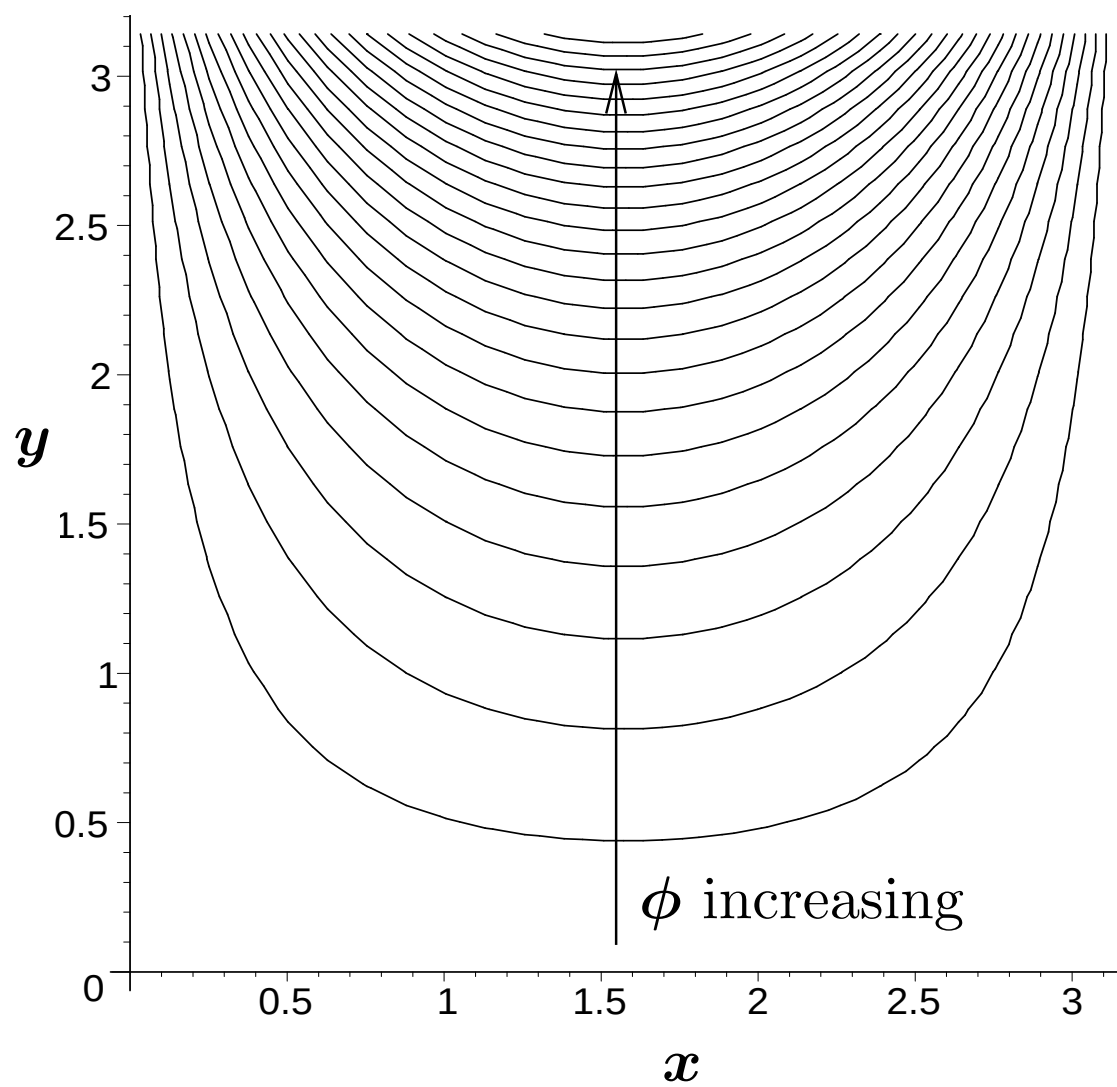


Figure 9:

Exercise. How would you calculate $\phi(\mathbf{x}, \mathbf{y})$ if instead of using the boundary data (82) and (83) you were given the boundary data

$$\phi(x, 0) = F(x), \quad \phi(x, \pi) = G(x), \quad \phi(0, y) = J(y), \quad \phi(\pi, y) = K(y)?$$

Example 5 Laplace's equation in polar coordinates (r, θ) is

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} = 0. \quad (89)$$

Find the a solution for $\phi(r, \theta)$ inside the unit disc where ϕ satisfies the boundary condition

$$\phi(1, \theta) = \begin{cases} 1 & \text{for } 0 < \theta < \pi/2, \\ -1 & \text{for } \pi/2 < \theta < \pi, \\ 1 & \text{for } \pi < \theta < 3\pi/2, \\ -1 & \text{for } 3\pi/2 < \theta < 2\pi. \end{cases} \quad (90)$$

$$\phi(0, \theta) \quad \text{is bounded} \quad (91)$$

$$\phi(r, \theta) \quad 2\pi\text{-periodic in } \theta \quad (92)$$

We separate variables and look for solutions to (89) of the form

$\phi = f(r)g(\theta)$ this gives

$$\begin{aligned} & \left(f''(r) + \frac{f'(r)}{r} \right) g(\theta) + \frac{f(r)}{r^2} g''(\theta) = 0 \\ \implies & g''(\theta) + \Lambda g(\theta) = 0 \end{aligned} \quad (93)$$

$$\text{and} \quad f''(r) + \frac{f'(r)}{r} - \Lambda \frac{f(r)}{r^2} = 0. \quad (94)$$

Requiring the solution to be periodic (92) gives

$$g(\theta + 2\pi) = g(\theta).$$

This, together with (93) comprises an eigenvalue problem for g and Λ . The only 2π - periodic solutions to (93) occur when $\Lambda = n^2$ (where n is an integer) and are

$$g(\theta) = A \cos(n\theta) + B \sin(n\theta).$$

The corresponding solutions to (94) are r^n and r^{-n} . Assuming n is a positive integer allows us to rule out the behaviour r^{-n} as this

leads to an unbounded solution at $\mathbf{r} = \mathbf{0}$ (and thus violates (91)). The general solution to the homogeneous problem comprising (89), (91) and (92) is thus

$$\phi(r, \theta) = \frac{A_0}{2} + \sum_{n=1}^{\infty} r^n (A_n \cos(n\theta) + B_n \sin(n\theta)). \quad (95)$$

In order to determine the coefficients $\{A_n\}$ and $\{B_n\}$ we apply the boundary condition (90) to get

$$\frac{A_0}{2} + \sum_{n=1}^{\infty} (A_n \cos(n\theta) + B_n \sin(n\theta)) = \begin{cases} 1, & 0 < \theta < \pi/2, \\ -1, & \pi/2 < \theta < \pi, \\ 1, & \pi < \theta < 3\pi/2, \\ -1, & 3\pi/2 < \theta < 2\pi. \end{cases}$$

Since the right hand side gives rise to an odd function in θ it follows that $A_n = 0$ for all n . The B_n are determined in the usual

fashion so that

$$\begin{aligned}
 B_n &= \frac{2}{\pi} \left[\int_0^{\pi/2} \sin(n\theta) d\theta - \int_{\pi/2}^{\pi} \sin(n\theta) d\theta \right], \\
 &= \frac{2}{\pi} \left\{ \left[-\frac{\cos(n\theta)}{n} \right]_0^{\pi/2} + \left[\frac{\cos(n\theta)}{n} \right]_{\pi/2}^{\pi} \right\}, \\
 \implies B_n &= \frac{2}{\pi} \left(\frac{1 + (-1)^n - 2 \cos(n\pi/2)}{n} \right).
 \end{aligned}$$

Hence

$$\phi(r, \theta) = \sum_{n=1}^{\infty} \frac{2r^n}{\pi} \left(\frac{1 + (-1)^n - 2 \cos(n\pi/2)}{n} \right) \sin(n\theta)$$

6.3 The general solution to the wave equation (51)

We make the following change of variables in (51):

$$\begin{aligned}\xi &= x - kt, & \eta &= x + kt \\ \Rightarrow \frac{\partial}{\partial x} &= \frac{\partial}{\partial \xi} + \frac{\partial}{\partial \eta} & \frac{\partial}{\partial t} &= k \left(\frac{\partial}{\partial \eta} - \frac{\partial}{\partial \xi} \right),\end{aligned}$$

so that it becomes

$$k^2 (u_{\eta\eta} - 2u_{\xi\eta} + u_{\xi\xi}) = c^2 (u_{\eta\eta} + 2u_{\xi\eta} + u_{\xi\xi}).$$

In particular if $k = c$ the equation reduces to

$$u_{\xi\eta} = 0.$$

This is easily integrated to give the general solution to the wave equation

$$u = f(\xi) + g(\eta) = f(x - ct) + g(x + ct), \quad (96)$$

where f and g are arbitrary functions.

6.4 The use of Legendre polynomials for PDEs involving the Laplacian in spherical polars

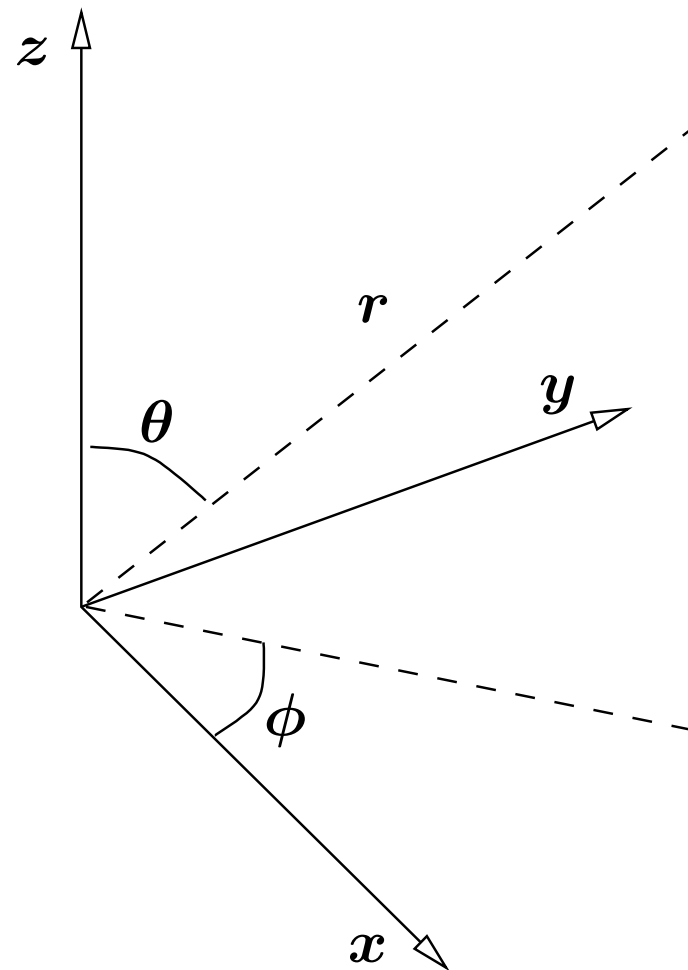
In terms of ‘spherical’ polar coordinates (r, θ, ϕ) , where $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$ and $z = r \cos \theta$ the Laplacian is given by

$$\nabla^2 \Psi = \left(\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \Psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \Psi}{\partial \phi^2} \right).$$

In this section we look for solutions to PDEs involving the Laplacian, by using the separation of variables, in spherical polars. A sketch of this coordinate system is given in figure 10.

6.4.1 Separation of variables for Laplace’s equation

We look for solutions to Laplace’s equation in spherical coordinates, with rotational symmetry (*i.e.* no ϕ dependence), so that



SPHERICAL POLARS

Figure 10: Spherical polar coordinates.

$$\left(\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \Psi}{\partial \theta} \right) \right) = 0, \quad (97)$$

$$\Psi \rightarrow 0, \quad \text{as } r \rightarrow \infty \quad (98)$$

by using separation of variables

$$\Psi(r, \theta) = F(r)G(\theta).$$

Substituting this ansatz into (97) and multiplying by $r^2/(F(r)G(\theta))$ leads to

$$\frac{1}{F(r)} \frac{d}{dr} \left(r^2 \frac{dF(r)}{dr} \right) + \frac{1}{G(\theta) \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dG(\theta)}{d\theta} \right) = 0. \quad (99)$$

In order to satisfy (99) we require

$$r^2 \frac{d^2 F}{dr^2} + 2r \frac{dF}{dr} - \Lambda F = 0, \quad (100)$$

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dG}{d\theta} \right) + \Lambda G = 0. \quad (101)$$

where Λ is an arbitrary constant. Equation (100) is an Euler equation and has solutions of the form $F = r^n$ where n satisfies the quadratic $n(n + 1) = \Lambda$. Henceforth we write $\Lambda = n(n + 1)$. The substitution $w = \cos \theta$ transforms (101) into the Legendre equation

$$(1 - w^2) \frac{d^2 G}{dw^2} - 2w \frac{dG}{dw} + n(n + 1)G = 0.$$

The general solution to which is

$$G(w) = AP_n(w) + BP_n(-w),$$

for n non-integer and

$$G(w) = AP_n(w) + BQ_n(w),$$

for n integer. Assuming the problem we are looking at includes the

poles $\theta = 0$ and $\theta = \pi$, where $w = 1$ and $w = -1$ respectively, we must choose n integer and B , the coefficient of Q_n , equal to zero in order to avoid singularities in G . Recall $Q_n(w)$ and $P_n(w)$ (for n non integer) have singularities at $w = -1$. The variable separable solutions to (97) are therefore

$$\Psi(r, \theta) = Ar^n P_n(\cos \theta) \quad \text{for } n \text{ integer}$$

Using the principle of superposition the most general solution to (97), without singularities at $\theta = 0$ and $\theta = \pi$, is

$$\Psi(r, \theta) = \left[\sum_{n=-\infty}^{-1} A_n r^n P_n(\cos \theta) \right] + \left[\sum_{n=0}^{\infty} A_n r^n P_n(\cos \theta) \right] \quad (102)$$

In practice we shall either require to solve (97) on the inside of the sphere, in which case we must set $A_n = 0$ in the first term in (102) to avoid singularities at the origin, or require to solve (97) on the outside of the sphere, in which case we must set $A_n = 0$ in the

second term in (102) to satisfy the condition $\Psi \rightarrow 0$ as $r \rightarrow \infty$.

Fourier Legendre Series. Any piecewise continuous function $g(\theta)$ defined between $\theta = 0$ and $\theta = \pi$ can be represented as a Fourier Legendre Series

$$g(\theta) = \sum_{n=0}^{\infty} \alpha_n P_n(\cos \theta). \quad (103)$$

This result follows from the equivalent result on Fourier series which says we can write $g(\theta) = \sum_{n=0}^{\infty} \beta_n \cos(n\theta)$ for some $\{\beta_n\}$. We can calculate the coefficients in the Fourier Legendre series by using the orthogonality relations for Legendre polynomials (see section on Legendre functions)

$$\int_{-1}^1 P_n(w) P_m(w) dw = \frac{2\delta_{mn}}{2m+1} \quad \text{for } m \geq 0, \quad n \geq 0. \quad (104)$$

Multiplying (103) by $P_m(\cos \theta) \sin \theta$ and integrating between $\theta = 0$ and $\theta = \pi$ gives

$$\int_0^\pi g(\theta) P_m(\cos \theta) \sin \theta d\theta = \sum_{n=0}^{\infty} \alpha_n \int_0^\pi P_n(\cos \theta) P_m(\cos \theta) \sin \theta d\theta.$$

By making the substitution $w = \cos \theta$ we obtain

$$\begin{aligned} \sum_{n=0}^{\infty} \alpha_n \int_{-1}^1 P_n(w) P_m(w) dw &= \int_{-1}^1 g(\arccos w) P_m(w) dw, \\ \implies \alpha_m &= \frac{2m+1}{2} \int_{-1}^1 g(\arccos w) P_m(w) dw. \end{aligned} \quad (105)$$

Example 5. Calculate the electric potential Ψ inside and outside of a spherical capacitor comprised of the two hemispheres $0 < \theta < \pi/2, r = 1$ and $\pi > \theta > \pi/2, r = 1$ given that the top hemisphere has potential $\Psi = 1$ while the bottom one has

potential $\Psi = 0$ and that Ψ satisfies Laplace's equation (97) and $\Psi \rightarrow 0$ as $r \rightarrow \infty$.

Inside the sphere $r < 1$ the general (nonsingular) solution to Laplace's equations is

$$\Psi(r, \theta) = \left[\sum_{n=0}^{\infty} A_n r^n P_n(\cos \theta) \right]. \quad (106)$$

while outside the sphere the general solution, which tends to zero as $r \rightarrow \infty$, is

$$\Psi(r, \theta) = \left[\sum_{n=1}^{\infty} B_n r^{-n} P_{-n}(\cos \theta) \right].$$

which we can rewrite as

$$\Psi(r, \theta) = \left[\sum_{n=1}^{\infty} B_n r^{-n} P_{n-1}(\cos \theta) \right]. \quad (107)$$

by using the relation $P_{-n}(w) = P_{n-1}(w)$. The boundary conditions

$$\Psi|_{r=1, \theta < \pi/2} = 1, \quad \Psi|_{r=1, \theta > \pi/2} = 0,$$

therefore imply

$$\left[\sum_{n=0}^{\infty} A_n P_n(\cos \theta) \right] = \begin{cases} 1 & \text{for } 0 \leq \theta < \frac{\pi}{2} \\ 0 & \text{for } \frac{\pi}{2} \leq \theta \leq \pi \end{cases} \quad (108)$$

$$\left[\sum_{n=1}^{\infty} B_n P_{n-1}(\cos \theta) \right] = \begin{cases} 1 & \text{for } 0 \leq \theta < \frac{\pi}{2} \\ 0 & \text{for } \frac{\pi}{2} \leq \theta \leq \pi \end{cases} \quad (109)$$

We calculate the coefficients $\{A_n\}$ and $\{B_n\}$ from the results on

Fourier Legendre series given in (103) and (105).

$$A_m = \frac{2m+1}{2} \int_{-1}^1 g(\arccos(w)) P_m(w) dw,$$

where

$$g(\theta) = \begin{cases} 1 & 0 \leq \theta < \frac{\pi}{2} \\ 0 & \frac{\pi}{2} \leq \theta \leq \pi \end{cases} \implies g(\arccos(w)) = \begin{cases} 0 & -1 \leq w < 0 \\ 1 & 0 \leq w \leq 1 \end{cases}$$

Hence

$$A_m = \frac{2m+1}{2} \int_0^1 P_m(w) dw.$$

A similar argument can be used to find B_m and it gives

$$B_m = \frac{2m-1}{2} \int_0^1 P_{m-1}(w) dw \implies B_m = A_{m-1} \quad \text{for } m \geq 1$$

The first few terms in this series may be calculated explicitly from the first few Legendre Polynomials

$$A_0 = \frac{1}{2} \int_0^1 dw = \frac{1}{2} \quad A_1 = \frac{3}{2} \int_0^1 w dw = \frac{3}{4} \quad A_2 = \frac{5}{4} \int_0^1 (3w^2 - 1) dw = 0.$$

The contours of Ψ (spacings **0.1** between $\Psi = 0$ and $\Psi = 1$), both within and outside the sphere, are plotted in figure 11 using 31 terms in its series.

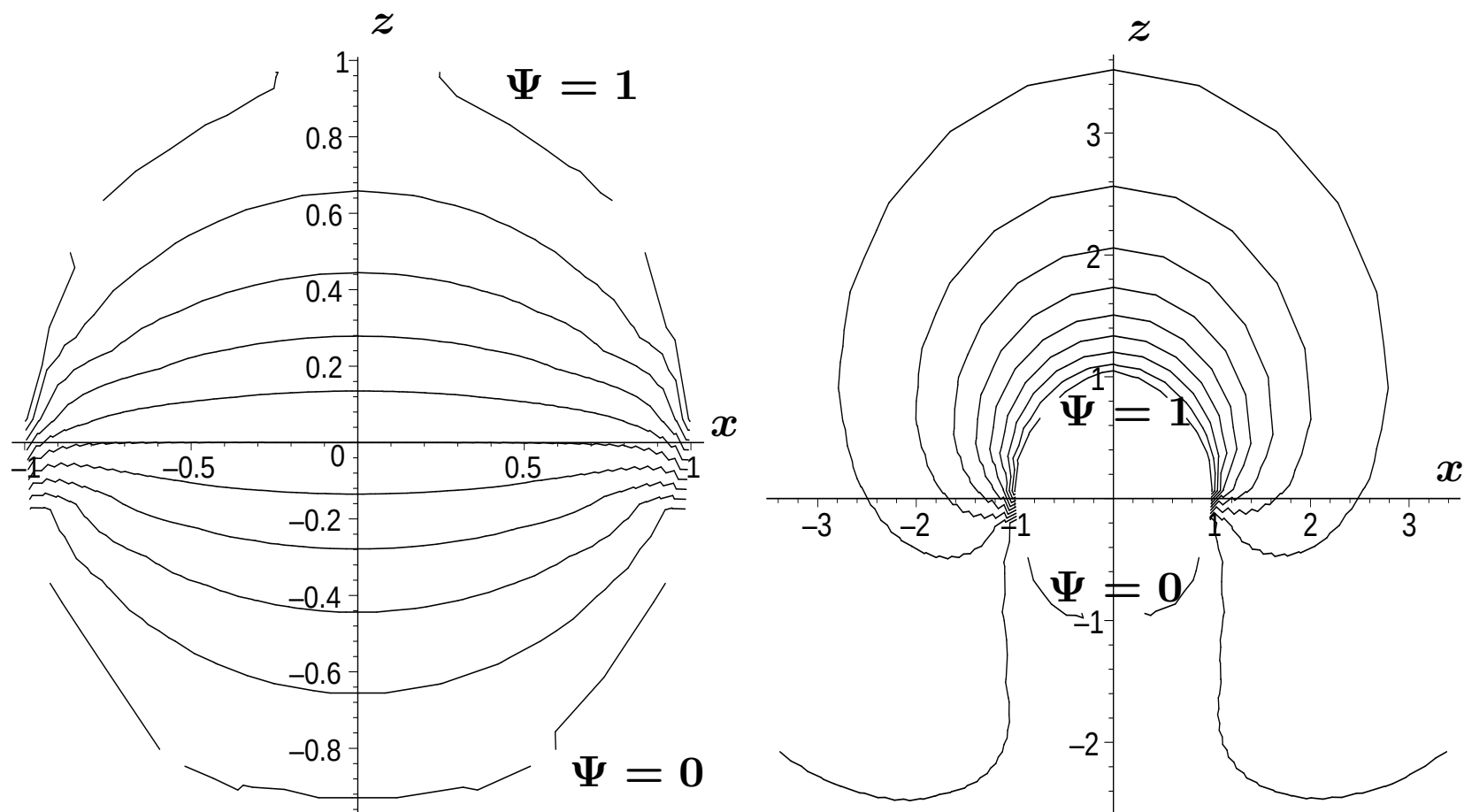


Figure 11:

7 Fourier Transforms

In this section we look at Fourier transforms. These are an extension of Fourier series to non-periodic functions $f(x)$ which have range $x \in (-\infty, \infty)$, but which decay to zero at infinity. The relation between Fourier transforms and Fourier series becomes clearer if we regard such functions as being periodic over the infinite interval $(-\infty, \infty)$. In practice Fourier transforms are commonly used to solve certain classes of linear PDEs and integral equations and to analyse data with the aim of predicting trends.

7.1 Derivation of the Fourier Transform

Consider the Fourier series for the $2L$ periodic function f

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L}x\right) + b_n \sin\left(\frac{n\pi}{L}x\right) \quad (110)$$

$$\text{where } a_m = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{m\pi}{L}x\right) dx, \quad (111)$$

$$\text{and } b_m = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{m\pi}{L}x\right) dx. \quad (112)$$

We now note that a_n is even in n and b_n is odd in n so that both $a_n \cos((n\pi/L)x)$ and $b_n \sin((n\pi/L)x)$ are even in n . It follows that, where we write

$$k_n = \Delta_k n, \quad \Delta_k = \frac{\pi}{L}. \quad (113)$$

we can rewrite (110) as

$$f(x) = \frac{1}{2} \sum_{n=-\infty}^{\infty} a_n \cos(k_n x) + b_n \sin(k_n x),$$

define

$$\begin{aligned} \alpha(k_n) &= \frac{\pi}{\Delta_k} a_n = \int_{-L}^L f(x) \cos(k_n x) dx, \\ \beta(k_n) &= \frac{\pi}{\Delta_k} b_n = \int_{-L}^L f(x) \sin(k_n x) dx, \end{aligned} \tag{114}$$

and rewrite $f(x)$ in terms of these new functions

$$f(x) = \frac{\Delta_k}{2\pi} \left[\sum_{n=-\infty}^{\infty} \alpha(k_n) \cos(k_n x) + \beta(k_n) \sin(k_n x) \right]. \tag{115}$$

We now take the limit $L \rightarrow \infty$ (note this means that $f(x)$ is no longer required to be periodic). As we do so the spacing between

the discrete values of $\mathbf{k}_n$ tends to zero; that is $\Delta_{\mathbf{k}} \rightarrow \mathbf{0}$. In this limit (114) and (115) give

$$\begin{aligned}\alpha(\mathbf{k}) &= \int_{-\infty}^{\infty} f(\mathbf{x}) \cos(\mathbf{k}\mathbf{x}) d\mathbf{x}, \\ \beta(\mathbf{k}) &= \int_{-\infty}^{\infty} f(\mathbf{x}) \sin(\mathbf{k}\mathbf{x}) d\mathbf{x},\end{aligned}\tag{116}$$

$$f(\mathbf{x}) = \frac{1}{2\pi} \int_{-\infty}^{\infty} [\alpha(\mathbf{k}) \cos(\mathbf{k}\mathbf{x}) + \beta(\mathbf{k}) \sin(\mathbf{k}\mathbf{x})] d\mathbf{k},\tag{117}$$

The first two of these integrals (116) are the Fourier transform of the function $f(\mathbf{x})$ while the second is termed the inverse Fourier transform. Roughly speaking, for this result to hold, we require f to be a piecewise continuous function of $\mathbf{x}$ and $f \rightarrow \mathbf{0}$ as $\mathbf{x} \rightarrow \pm\infty$.

It is more usual to write the Fourier transform and its inversion (116)-(117) in terms of complex numbers. We can do this by defining a new (complex) function by $\hat{f}(\mathbf{k}) = \alpha(\mathbf{k}) + i\beta(\mathbf{k})$. In

terms of $\hat{f}$ we can rewrite (116)-(117) as

$$\hat{f}(k) = \int_{-\infty}^{\infty} f(x)(\cos(kx) + i \sin(kx))dx = \int_{-\infty}^{\infty} f(x)e^{ikx}dx.$$

$$f(x) = \frac{1}{2\pi} \left[\int_{-\infty}^{\infty} \hat{f}(k)(\cos(kx) - i \sin(kx))dk \right. \\ \left. - i \int_{-\infty}^{\infty} (\beta(k) \cos(kx) - \alpha(k) \sin(kx))dk \right].$$

We can simplify the latter by noting that $\beta(k) \cos(kx)$ and $\alpha(k) \sin(kx)$ are both odd in k and so the integral premultiplied by i vanishes. It follows that the Fourier transform and its inversion are

$$\hat{f}(k) = \int_{-\infty}^{\infty} e^{ikx} f(x) dx, \quad (118)$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ikx} \hat{f}(k) dk. \quad (119)$$

Beware this is not the only definition of a Fourier transform.

Some books use the transformed variable $\hat{f}/\sqrt{2\pi}$, but call it $\hat{f}$. It is also common to see the Fourier transform of the function f to be written in operator notation as $\mathcal{F}(f)$. Thus $\hat{f}(k) = \mathcal{F}(f(x))$ and $f(x) = \mathcal{F}^{-1}(\hat{f}(k))$.

7.2 Fourier cosine transform and Fourier sine transform

If $f(x)$ is an even function of x then $\beta(k) = 0$ and we may rewrite (116) and (117) as

$$\alpha(k) = 2 \int_0^{\infty} f(x) \cos(kx) dx, \quad f(x) = \frac{1}{\pi} \int_0^{\infty} \alpha(k) \cos(kx) dk.$$

If we then make the substitution $\alpha(k) = 2F_c(k)$ we retrieve the Fourier cosine transform

$$F_c(k) = \int_0^{\infty} f(x) \cos(kx) dx, \quad (120)$$

$$f(x) = \frac{2}{\pi} \int_0^{\infty} F_c(k) \cos(kx) dk. \quad (121)$$

Alternatively if $f(x)$ is an odd function of x then $\alpha(k) = 0$ and

we may rewrite (116) and (117) as

$$\beta(k) = 2 \int_0^\infty f(x) \sin(kx) dx, \quad f(x) = \frac{1}{\pi} \int_0^\infty \beta(k) \sin(kx) dk.$$

Making the substitution $\beta(k) = 2F_s(k)$ we retrieve the Fourier sine transform

$$F_s(k) = \int_0^\infty f(x) \sin(kx) dx, \quad (122)$$

$$f(x) = \frac{2}{\pi} \int_0^\infty F_s(k) \sin(kx) dk. \quad (123)$$

Note that the Fourier cosine and sine transforms relate to the Fourier transform via the formulae

$$F_c(k) = \frac{1}{2} \hat{f}(k), \quad \text{for } f(x) = f(-x),$$

$$F_s(k) = \frac{1}{2i} \hat{f}(k), \quad \text{for } f(x) = -f(-x).$$

The use of the Fourier cosine and sine transform is not so much in dealing with odd and even function but in solving problems in which f is only defined in the interval $x \in [0, \infty)$. In this instance since f is undefined in $x \in (-\infty, 0)$ we can always extend it into this interval to ensure that it is odd or even as required. You may also come across tables of Fourier cosine and sine transforms and it is useful to be able to relate these to the full Fourier transform.

7.3 Important properties of the Fourier transform.

Linearity The Fourier transform is linear *i.e.*

$$\mathcal{F}(af(x) + bg(x)) = a\hat{f}(k) + b\hat{g}(k).$$

This is easily verified on referring to (116)-(117).

Fourier transform of derivatives.

$$\begin{aligned}\mathcal{F}\left(\frac{df}{dx}\right) &= \int_{-\infty}^{\infty} \frac{df}{dx} e^{ikx} dx = \int_{-\infty}^{\infty} \frac{d}{dx} (e^{ikx} f) - ik e^{ikx} f dx \\ &= [e^{ikx} f]_{-\infty}^{\infty} - ik \hat{f}(k).\end{aligned}$$

The requirement that $f \rightarrow 0$ as $x \rightarrow \pm\infty$ implies that $[e^{ikx} f]_{-\infty}^{\infty} = 0$ and hence that

$$\mathcal{F}\left(\frac{df}{dx}\right) = -ik \hat{f}(k). \quad (124)$$

Further derivatives may be easily calculated as follows:

$$\mathcal{F}\left(\frac{d^2 f}{dx^2}\right) = -ik \mathcal{F}\left(\frac{df}{dx}\right) = -k^2 \hat{f}(k).$$

This leads to the general formula for the n 'th derivative

$$\mathcal{F}\left(\frac{d^n f}{dx^n}\right) = (-ik)^n \hat{f}(k). \quad (125)$$

Convolution. The convolution $f * g$ of two functions f and g is defined by

$$(f * g)(x) = \int_{-\infty}^{\infty} f(p)g(x - p)dp = \int_{-\infty}^{\infty} f(x - p)g(x)dp, \quad (126)$$

(exercise: show that two integrals above are identical). The Fourier transform of the convolution $f * g$ is

$$\mathcal{F}((f * g)(x)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{ikx} f(p)g(x - p)dpdx.$$

Assuming we are allowed to **swap** the order of integration we can make the substitution $x = p + q$ to find

$$\begin{aligned} \mathcal{F}((f * g)(x)) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{ik(p+q)} f(p)g(q)dpdq \\ &= \left(\int_{-\infty}^{\infty} e^{ikp} f(p)dp \right) \left(\int_{-\infty}^{\infty} e^{ikq} g(q)dq \right), \end{aligned}$$

$$\implies \mathcal{F}((f * g)(x)) = \hat{f}(k)\hat{g}(k). \quad (127)$$

This result gives an efficient method of solving integral equations involving convolutions of the type (126).

Shifting theorem.

$$\mathcal{F}(f(x - a)) = \int_{-\infty}^{\infty} f(x - a)e^{ikx} dx = \int_{-\infty}^{\infty} f(x')e^{ikx'} e^{ika} dx'$$

where we make the change of variables $x' = x - a$. It follows that

$$\mathcal{F}(f(x - a)) = e^{ika} \hat{f}(k). \quad (128)$$

7.4 Calculating Fourier Transforms.

In practice there are few functions that can be Fourier transformed without the use of techniques from complex analysis (these are

taught in G1BCOF). This means that you will usually have to refer to look-up tables of Fourier Transforms when you come to use this technique. There are, however, a few functions which we can transform explicitly. The most important of these are listed below.

Fourier transform of multiples of x . Note that

$$\frac{d}{dk} \int_{-\infty}^{\infty} e^{ikx} f(x) dx = i \int_{-\infty}^{\infty} x e^{ikx} f(x) dx.$$

It follows that

$$\mathcal{F}(xf(x)) = -i \frac{d\hat{f}(k)}{dk}. \quad (129)$$

This is easily generalised to higher powers of x

$$\mathcal{F}(x^m f(x)) = (-i)^m \frac{d^m \hat{f}(k)}{dk^m}. \quad (130)$$

Fourier transforms of exponentials. The function $e^{-\alpha|x|}$ is an even function in x which decays as $x \rightarrow \pm\infty$ for $\alpha > 0$; thus

$$\begin{aligned}
 \mathcal{F} \left(e^{-\alpha|x|} \right) &= \int_{-\infty}^{\infty} e^{ikx - \alpha|x|} dx \\
 &= \int_{-\infty}^0 e^{(ik+\alpha)x} dx + \int_0^{\infty} e^{(ik-\alpha)x} dx \\
 &= \left[\frac{e^{(ik+\alpha)x}}{ik+\alpha} \right]_{-\infty}^0 + \left[\frac{e^{(ik-\alpha)x}}{ik-\alpha} \right]_0^{\infty} . \\
 \implies \mathcal{F} \left(e^{-\alpha|x|} \right) &= \frac{2\alpha}{\alpha^2 + k^2} . \tag{131}
 \end{aligned}$$

A similar result holds when we reverse the process and look for the inverse Fourier transform of $e^{-a|k|}$ this gives

$$\mathcal{F} \left(\frac{1}{x^2 + a^2} \right) = \frac{\pi}{a} e^{-a|k|} \quad (132)$$

The function $\exp(-\alpha x^2)$ decays for $\alpha > 0$ and has Fourier transform given by

$$\begin{aligned} \mathcal{F} \left(e^{-\alpha x^2} \right) &= \int_{-\infty}^{\infty} \exp(-\alpha x^2 + ikx) dx \\ &= \int_{-\infty}^{\infty} \exp \left(-\alpha \left(x - \frac{ik}{2\alpha} \right)^2 - \frac{k^2}{4\alpha} \right) dx \\ &= \exp \left(-\frac{k^2}{4\alpha} \right) \int_{-\infty}^{\infty} \exp(-\alpha z^2) dz \\ &= \frac{1}{\sqrt{\alpha}} \exp \left(-\frac{k^2}{4\alpha} \right) \int_{-\infty}^{\infty} \exp(-\tilde{z}^2) d\tilde{z}. \end{aligned}$$

It follows that

$$\mathcal{F}\left(e^{-\alpha x^2}\right) = \sqrt{\frac{\pi}{\alpha}} \exp\left(-\frac{k^2}{4\alpha}\right). \quad (133)$$

7.5 Examples

Example 1. Use the Fourier transform to find the solution to the wave equation

$$u_{tt} = c^2 u_{xx} \quad (134)$$

subject to the initial conditions

$$u|_{t=0} = f(x), \quad \left.\frac{\partial u}{\partial t}\right|_{t=0} = 0. \quad (135)$$

Taking the Fourier transform of the wave equation gives

$$\hat{u}_{tt} = -(ck)^2 \hat{u}.$$

This has solution

$$\hat{u} = A(k) \cos((ck)t) + B(k) \sin((ck)t).$$

The Fourier transform of the initial conditions is

$$\hat{u}|_{t=0} = \hat{f}(k), \quad \left. \frac{\partial \hat{u}}{\partial t} \right|_{t=0} = 0.$$

Applying these initial conditions to the solution for $\hat{u}$ gives $A = \hat{f}(k)$ and $B = 0$ so that

$$\hat{u}(k, t) = \hat{f}(k) \cos((ck)t) = \hat{f}(k) \frac{e^{ikct} + e^{-ikct}}{2}.$$

We can invert this using the shifting theorem (128) and linearity of the Fourier transform. The former gives

$$\mathcal{F}^{-1} \left(\hat{f}(k) e^{ikct} \right) = f(x - ct), \quad \mathcal{F}^{-1} \left(\hat{f}(k) e^{-ikct} \right) = f(x + ct).$$

while the latter implies

$$u(x, t) = \mathcal{F}^{-1}(\hat{u}(k, t)) = \frac{1}{2} (f(x - ct) + f(x + ct)).$$

Example 2. Use the Fourier transform to solve the heat equation

$$u_t = u_{xx} \tag{136}$$

in an infinite spatial domain subject to the initial and boundary conditions

$$u|_{t=0} = f(x), \quad \left. \begin{array}{l} u \rightarrow 0 \\ u_x \rightarrow 0 \end{array} \right\} \quad \text{as } |x| \rightarrow \infty. \tag{137}$$

Taking the Fourier transform of (136) and the initial conditions

(137a) gives

$$\hat{u}_t = -k^2 \hat{u}, \quad \hat{u}|_{t=0} = \hat{f}(k).$$

Solving the former and applying the initial condition provided by the latter gives

$$\hat{u}(k, t) = \hat{f}(k) \exp(-k^2 t). \quad (138)$$

We invert by noting that $\hat{u}$ is comprised of two functions of k multiplied together; if we can find the functions of x and t which transform to these functions of k we can use the convolution theorem to invert for $u(x, t)$. Recall formula (133)

$$\mathcal{F} \left(e^{-\alpha x^2} \right) = \sqrt{\frac{\pi}{\alpha}} \exp \left(-\frac{k^2}{4\alpha} \right).$$

Taking $\alpha = 1/(4t)$ gives the required result

$$\mathcal{F}^{-1}(\exp(-k^2 t)) = \frac{1}{2\pi^{1/2} t^{1/2}} \exp\left(-\frac{x^2}{4t}\right).$$

Using the inverse of the convolution theorem (127) on (138) gives

$$\begin{aligned} u(x, t) &= \left(f(x) * \frac{1}{2\pi^{1/2} t^{1/2}} \exp\left(-\frac{x^2}{4t}\right) \right) \\ &= \frac{1}{2\pi^{1/2} t^{1/2}} \int_{-\infty}^{\infty} f(p) \exp\left(-\frac{(x-p)^2}{4t}\right) dp. \end{aligned}$$

This is the ‘Green’s’ function formulation of the solution to the heat equation in an infinite domain.

8 Laplace Transforms

Laplace transforms can be used as a tool for solving linear differential equations, certain classes of integral equations and linear delay differential equations.

Definition of Laplace transform. The function $f(t)$ with range $t \in [0, \infty)$ has Laplace transform $\bar{f}(p)$ defined by

$$\bar{f}(p) = \mathcal{L}(f) = \int_0^{\infty} e^{-pt} f(t) dt. \quad (139)$$

If we define $f(t) = 0$ for $t < 0$ then the Fourier transform of $f(t)$ is given by $\mathcal{F}(f) = \bar{f}(-ik)$. It is usual to use this fact to derive the inversion integral for the Laplace transform, which, it turns

out, runs parallel to the imaginary axis and is given by

$$f(t) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} e^{pt} \bar{f}(p) dp \quad t \geq 0, \quad (140)$$

where γ is any sufficiently large positive constant. However you have not met enough complex analysis to perform such integrals. In the context of this module it is enough to know that such a transform exists and that given $\bar{f}(p)$ (with suitable behaviour) it is possible to find its unique inverse $f(t)$; *i.e.*

$$f(t) = \mathcal{L}^{-1}(\bar{f}).$$

Thus no two non-identical functions have the same Laplace transform. In practice you will be able to invert the Laplace transform using tables.

Existence. Provided the function $f(t)$ is piecewise continuous and has growth at infinity bounded by an exponential

$$|f(t)| < Me^{\mu t} \quad \text{for some } \mu \text{ and } M$$

then the integral in (139) exists for $p > \mu$ and hence the Laplace transform of $f(t)$ also exists (even though it might not exist for values of $p \leq \mu$). This might appear a peculiar definition of existence, but given the nature of the inversion contour (140), all we require in order to invert $\bar{f}(p)$ is that it is defined in the range $p \in (\mu, \infty)$ for some finite value μ .

8.1 Properties of the Laplace Transform

Linearity The Laplace transform is linear; that is

$$\mathcal{L}(\alpha f + \beta g) = \alpha \bar{f}(p) + \beta \bar{g}(p), \quad (141)$$

where α and β are arbitrary constants (exercise: show this from the definition (139)). An important consequence of linearity is that

$$\bar{h}(p) = \alpha \bar{f}(p) + \beta \bar{g}(p),$$

can be inverted by first inverting $\bar{f}(p)$ and $\bar{g}(p)$ separately, and then combining the results to obtain the answer

$$h(t) = \alpha \mathcal{L}^{-1}(\bar{f}) + \beta \mathcal{L}^{-1}(\bar{g}).$$

Laplace transform of derivatives. The Laplace transform of $\partial f / \partial t$ is

$$\begin{aligned} \mathcal{L} \left(\frac{\partial f}{\partial t} \right) &= \int_0^\infty \frac{\partial f}{\partial t} e^{-pt} dt = \int_0^\infty \frac{\partial}{\partial t} (e^{-pt} f) + p e^{-pt} f dt \\ &= [e^{-pt} f]_0^\infty + p \mathcal{L}(f) \\ \implies \mathcal{L} \left(\frac{\partial f}{\partial t} \right) &= p \bar{f}(p) - f|_{t=0}. \end{aligned} \tag{142}$$

It is easy to generalise this to higher derivatives simply by applying (142) an appropriate number of times; thus

$$\begin{aligned}\mathcal{L}\left(\frac{\partial^2 f}{\partial t^2}\right) &= p\mathcal{L}\left(\frac{\partial f}{\partial t}\right) - \frac{\partial f}{\partial t}\Big|_{t=0} \\ &= p^2 \bar{f}(p) - pf|_{t=0} - \frac{\partial f}{\partial t}\Big|_{t=0}.\end{aligned}$$

Laplace transforms of indefinite integrals.

$$\begin{aligned}\mathcal{L}\left(\int_0^t f(\tau)d\tau\right) &= \int_0^\infty \left[\int_0^t f(\tau)d\tau\right] e^{-pt} dt \\ &= \int_0^\infty \frac{d}{dt} \left(-\frac{e^{-pt}}{p} \left[\int_0^t f(\tau)d\tau\right]\right) + \frac{e^{-pt}}{p} f(t) dt \\ &\implies \mathcal{L}\left(\int_0^t f(\tau)d\tau\right) = \frac{1}{p} \mathcal{L}(f).\end{aligned}\tag{143}$$

Laplace transforms of convolutions. In the context of Laplace transforms the convolution of two functions f and g is defined by

$$(f * g)(t) = \int_0^t f(\tau)g(t - \tau)d\tau = \int_0^t f(t - u)g(u)du.$$

Note that this definition is different to the one used when discussing Fourier transforms. Taking the Laplace transform of $(f * g)$ gives

$$\mathcal{L}(f * g) = \int_{t=0}^{\infty} e^{-pt} \left[\int_{\tau=0}^t f(\tau)g(t - \tau)d\tau \right] dt, \quad (144)$$

where we integrate over τ first. Reversing the order of integration, which takes place over the shaded region in figure 12, gives

$$\begin{aligned} \mathcal{L}(f * g) &= \int_{\tau=0}^{\infty} f(\tau) \left[\int_{t=\tau}^{\infty} e^{-pt} g(t - \tau)dt \right] d\tau \\ &= \int_{\tau=0}^{\infty} f(\tau) \left[\int_{u=0}^{\infty} e^{-pu} e^{-p\tau} g(u)du \right] d\tau, \end{aligned}$$

on substituting $u = t - \tau$. It follows that

$$\mathcal{L}(f * g) = \mathcal{L}(f)\mathcal{L}(g) = \bar{f}(p)\bar{g}(p). \quad (145)$$

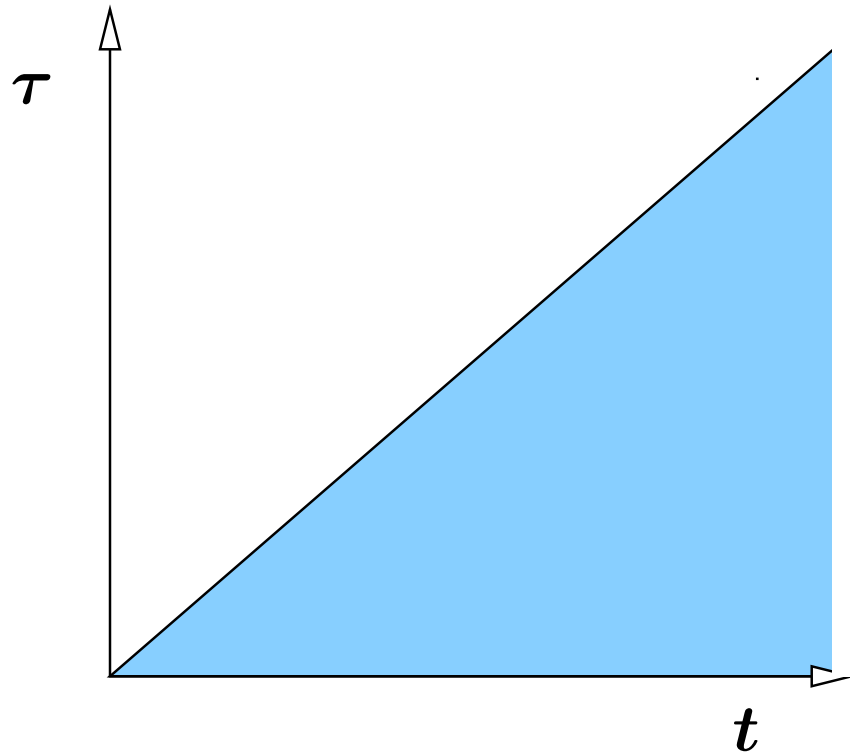


Figure 12: Area of integration in 144.

The shifting theorem 1. Consider

$$\begin{aligned}\mathcal{L}(f(t)e^{at}) &= \int_0^{\infty} e^{-(p-a)t} f(t) dt \\ \implies \mathcal{L}(f(t)e^{at}) &= \bar{f}(p-a)\end{aligned}\tag{146}$$

This result is called the shifting theorem, as the effect of multiplying $f(t)$ by e^{at} is to shift its Laplace transform by distance a . It is particularly useful when inverting Laplace transforms. Note, however, that if $\bar{f}(p)$ is only defined for values of $p > p_{crit}$ then $\mathcal{L}(f(t)e^{at})$ is only defined for values of $p > a + p_{crit}$.

The shifting theorem 2. The Heaviside function is defined as follows:

$$H(x) = \begin{cases} 0 & x < 0 \\ 1 & x > 0 \end{cases}\tag{147}$$

Taking the Laplace transform of $f(t - a)H(t - a)$ (where $a > 0$) gives

$$\begin{aligned}\mathcal{L}(f(t - a)H(t - a)) &= \int_a^\infty e^{-pt} f(t - a) dt \\ &= \int_0^\infty e^{-pu} e^{-pa} f(u) du\end{aligned}$$

where we substitute $u = t - a$. This implies

$$\mathcal{L}(f(t - a)H(t - a)) = e^{-pa} \mathcal{L}(f(t)) = e^{-pa} \bar{f}(p).$$

Laplace transform of a function multiplied by t .

Consider the derivative of $\bar{f}(p)$

$$\begin{aligned}\frac{d}{dp} \bar{f}(p) &= \frac{d}{dp} \int_0^\infty f(t) e^{-pt} dt \\ &= \int_0^\infty -t f(t) e^{-pt} dt = -\mathcal{L}(t f(t))\end{aligned}$$

$$\implies \mathcal{L}(tf(t)) = -\frac{d}{dp}\bar{f}(p)$$

Scaling of the independent variable. Taking the Laplace transform of $f(at)$ gives

$$\mathcal{L}(f(at)) = \int_0^\infty f(at)e^{-pt}dt = \frac{1}{a} \int_0^\infty f(u)e^{-(p/a)u}du$$

substituting $u = at$

$$\implies \mathcal{L}(f(at)) = \frac{1}{a}\bar{f}\left(\frac{p}{a}\right). \quad (148)$$

8.2 Laplace transforms of common functions.

Laplace transforms of powers of t . The Laplace transform of 1 is given by

$$\mathcal{L}(1) = \int_0^{\infty} e^{-pt} dt = \frac{1}{p} \quad (149)$$

We can calculate $\mathcal{L}(t^n)$ by using integration by parts; hence

$$\begin{aligned} \mathcal{L}(t^n) &= \int_0^{\infty} e^{-pt} t^n dt = \int_0^{\infty} \frac{d}{dt} \left(-\frac{e^{-pt}}{p} t^n \right) + \frac{e^{-pt}}{p} n t^{n-1} dt \\ &\implies \mathcal{L}(t^n) = \frac{n}{p} \mathcal{L}(t^{n-1}). \end{aligned}$$

It follows from the above and (149) that

$$\mathcal{L}(t^n) = \frac{n!}{p^{n+1}} \quad (150)$$

Laplace transforms of exponentials and trigonometric functions. The Laplace transform of e^{at} is given by

$$\mathcal{L}(e^{at}) = \frac{1}{p - a}. \quad (151)$$

It follows that the Laplace transform of $e^{i\omega t}$ is

$$\mathcal{L}(e^{i\omega t}) = \frac{1}{p - i\omega} = \frac{p}{p^2 + \omega^2} + i \frac{\omega}{p^2 + \omega^2}.$$

Taking real and imaginary parts of this expression leads to expressions for the Laplace transforms of $\cos(\omega t)$ and $\sin(\omega t)$

$$\mathcal{L}(\cos(\omega t)) = \frac{p}{p^2 + \omega^2}, \quad (152)$$

$$\mathcal{L}(\sin(\omega t)) = \frac{\omega}{p^2 + \omega^2}. \quad (153)$$

8.3 Uses of the Laplace transform

Example 1. Calculate the function $f(t)$ which is defined in terms of its Laplace transform as follows:

$$f(t) = \mathcal{L}^{-1} (\bar{f}(p)) = \mathcal{L}^{-1} \left(\frac{1}{p^3(1+p^2)} \right). \quad (154)$$

We first write this function of p using partial fractions

$$\bar{f}(p) = \frac{1}{p^3(1+p^2)} = \frac{\alpha_1 p + \alpha_0}{1+p^2} + \frac{\beta_3}{p^3} + \frac{\beta_2}{p^2} + \frac{\beta_1}{p}.$$

Recombining the right-hand side and equating powers of p with the left-hand side gives $\alpha_1 = 1$, $\beta_1 = -1$, $\beta_3 = 1$ and $\alpha_0 = \beta_2 = 0$ so that

$$\bar{f}(p) = \frac{p}{1+p^2} + \frac{1}{p^3} - \frac{1}{p}.$$

It is now simple to invert for $f(t)$ using the results on

trigonometric functions (152) and those on powers of t (150) to find

$$f(t) = \cos(t) + \frac{1}{2}t^2 - 1.$$

An alternative way of finding $f(t)$ is to use the convolution theorem directly on $\bar{f}(p)$, as expressed in (154); this approach yields

$$\begin{aligned} f(t) &= \mathcal{L}^{-1} \left(\frac{1}{p^3} \right) * \mathcal{L}^{-1} \left(\frac{1}{(1+p^2)} \right) = \left(\frac{t^2}{2} \right) * (\sin t) \\ \Rightarrow f(t) &= \frac{1}{2} \int_0^t \tau^2 \sin(t - \tau) d\tau \\ &= \frac{1}{2} \int_0^t \frac{d}{d\tau} (\tau^2 \cos(t - \tau) + 2\tau \sin(t - \tau)) \\ &\quad - 2 \sin(t - \tau) d\tau \\ \Rightarrow f(t) &= \frac{1}{2} \left[\tau^2 \cos(t - \tau) + 2\tau \sin(t - \tau) - 2 \cos(t - \tau) \right]_{\tau=0}^{\tau=t} \end{aligned}$$

$$= \cos(t) + \frac{1}{2}t^2 - 1.$$

Example 2. Laplace transforms are often used to solve linear ODEs with constant coefficients (there are many other methods available to solve such equations). In this example we solve

$$\frac{d^2 y}{dt^2} - 4\frac{dy}{dt} + 4y = e^{2t} \quad (155)$$

using Laplace transforms.

Taking the Laplace transform of (155) gives

$$\begin{aligned} p^2 \bar{y} - py(0) - \dot{y}(0) - 4p\bar{y} + 4y(0) + 4\bar{y} &= \frac{1}{p-2} \\ \implies (p-2)^2 \bar{y} &= py(0) + (\dot{y}(0) - 4y(0)) + \frac{1}{p-2}, \end{aligned}$$

$$\implies \bar{y} = \frac{p}{(p-2)^2}y(0) + \frac{(\dot{y}(0) - 4y(0))}{(p-2)^2} + \frac{1}{(p-2)^3}.$$

We now invert for $y(t)$ term by term by noting

$$\mathcal{L}^{-1} \left(\frac{1}{(p-2)^2} \right) = te^{2t} \quad \text{using (146) and (150)}$$

$$\begin{aligned} \text{But } \frac{p}{(p-2)^2} &= \frac{1}{p-2} + \frac{2}{(p-2)^2} \\ \implies \mathcal{L}^{-1} \left(\frac{p}{(p-2)^2} \right) &= e^{2t} + 2te^{2t} \end{aligned}$$

$$\mathcal{L}^{-1} \left(\frac{1}{(p-2)^3} \right) = \frac{t^2}{2}e^{2t} \quad \text{using (146) and (150)}$$

It follows that

$$\begin{aligned}y(t) &= y(0)(e^{2t} + 2e^{2t}t) + (\dot{y}(0) - 4y(0))te^{2t} + \frac{e^{2t}t^2}{2} \\ \implies y(t) &= y(0)e^{2t} + (\dot{y}(0) - 2y(0))te^{2t} + \frac{t^2e^{2t}}{2},\end{aligned}$$

Example 3. Show that the ODE

$$\frac{d^2y}{dt^2} + a\frac{dy}{dt} + by = r(t), \quad (156)$$

can always be written in the form

$$\frac{d^2y}{dt^2} - (\alpha + \beta)\frac{dy}{dt} + \alpha\beta y = r(t) \quad (157)$$

and then solve this equation using Laplace transforms in the case where α and β are distinct.

Writing $(\alpha + \beta) = -a$ and $\alpha\beta = b$ gives

$$\begin{aligned}\beta &= \frac{b}{\alpha} \implies \alpha + \frac{b}{\alpha} = -a \implies \alpha^2 + a\alpha + b = 0 \\ \implies \alpha &= \frac{-a \pm (a^2 - 4b)^{1/2}}{2}.\end{aligned}$$

It is thus clear that we can always find α and β so as to write (156) in the form (157). However in certain circumstances they are not real but, instead, form a complex conjugate pair. Taking the Laplace transform of (157) gives

$$\bar{y}(p) (p^2 - (\alpha + \beta)p + \alpha\beta) = \bar{r}(p) + py(0) + (\dot{y}(0) - (\alpha + \beta)y(0)).$$

Solving for $\bar{y}$ yields

$$\bar{y}(p) = \frac{\bar{r}(p)}{(p - \alpha)(p - \beta)} + \frac{\dot{y}(0) - (\alpha + \beta)y(0)}{(p - \alpha)(p - \beta)} + \frac{py(0)}{(p - \alpha)(p - \beta)}. \quad (158)$$

In order to invert this equation to find y it is helpful to use partial

fractions to derive the following expressions

$$\begin{aligned}\frac{1}{(p - \alpha)(p - \beta)} &= \frac{1}{\alpha - \beta} \left(\frac{1}{p - \alpha} - \frac{1}{p - \beta} \right), \\ \frac{p}{(p - \alpha)(p - \beta)} &= \frac{1}{\alpha - \beta} \left(\frac{\alpha}{p - \alpha} - \frac{\beta}{p - \beta} \right).\end{aligned}$$

Substituting these into (158) gives

$$\bar{y}(p) = \frac{1}{\alpha - \beta} \left[\frac{\bar{r}(p)}{(p - \alpha)} - \frac{\bar{r}(p)}{(p - \beta)} + \frac{\dot{y}(0) - \beta y(0)}{p - \alpha} - \frac{\dot{y}(0) - \alpha y(0)}{p - \beta} \right].$$

We can now invert this term by term. Recall that

$\mathcal{L}^{-1}(1/(p - \alpha)) = e^{\alpha t}$ and note that the first two terms can be inverted using the convolution theorem (145); hence

$$\begin{aligned}y(t) = \frac{1}{\alpha - \beta} & \left[(r * e^{\alpha t}) - (r * e^{\beta t}) + (\dot{y}(0) - \beta y(0))e^{\alpha t} \right. \\ & \left. - (\dot{y}(0) - \alpha y(0))e^{\beta t} \right]\end{aligned}$$

$$\begin{aligned} \Rightarrow y(t) = & \frac{1}{\alpha - \beta} \left[\int_0^t r(\tau) \left(e^{\alpha(t-\tau)} - e^{\beta(t-\tau)} \right) d\tau \right. \\ & \left. + (\dot{y}(0) - \beta y(0))e^{\alpha t} - (\dot{y}(0) - \alpha y(0))e^{\beta t} \right]. \end{aligned}$$

Example 5. Solve the integral equation

$$y(t) = 1 + \frac{3}{2} \int_0^t y(\tau) \sin(2(t - \tau)) d\tau, \quad (159)$$

using a Laplace transform.

Taking the Laplace transform of (159) gives

$$\bar{y}(p) = \frac{1}{p} + \frac{3}{2} \bar{y}(p) \frac{2}{p^2 + 4}$$

Rearranging this gives

$$\bar{y}(p) = \frac{p^2 + 4}{p(p^2 + 1)} = \frac{p}{p^2 + 1} + \frac{4}{p} \frac{1}{p^2 + 1}.$$

This can be inverted, using the results on trigonometric functions, (152) and (153), and on convolutions (145), to find

$$y(t) = \cos t + 4 \int_0^t \sin \tau d\tau = 4 - 3 \cos t.$$